

Analog 2

3.3V
Signal
Gnd
SDA
SCL

SDA
SCL
3.3V
Gnd

LED
Mark
Gnd

Up
Down
Left
Right
Enter
Gnd

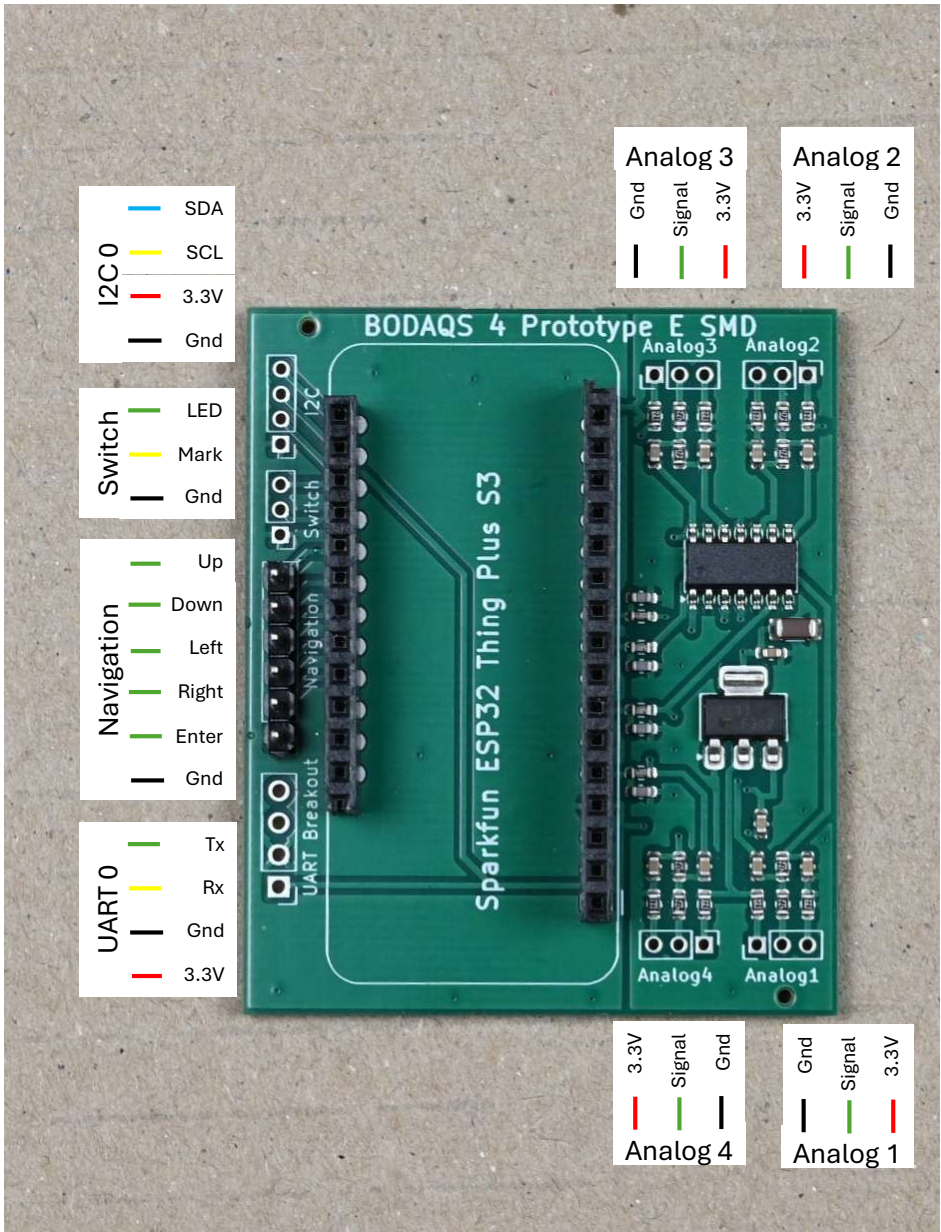
Tx
Rx
Gnd
3.3V

Analog 3
Gnd
Signal
3.3V

Analog 2
3.3V
Signal
Gnd

3.3V
Signal
Gnd
Analog 4

Gnd
Signal
3.3V
Analog 1



Inside (solder side)

I2C 0

Analog 3

Switch

Analog 2

Analog 1

Analog 4

Outside (connector side)

Analog 3

I2C 0

Analog 4

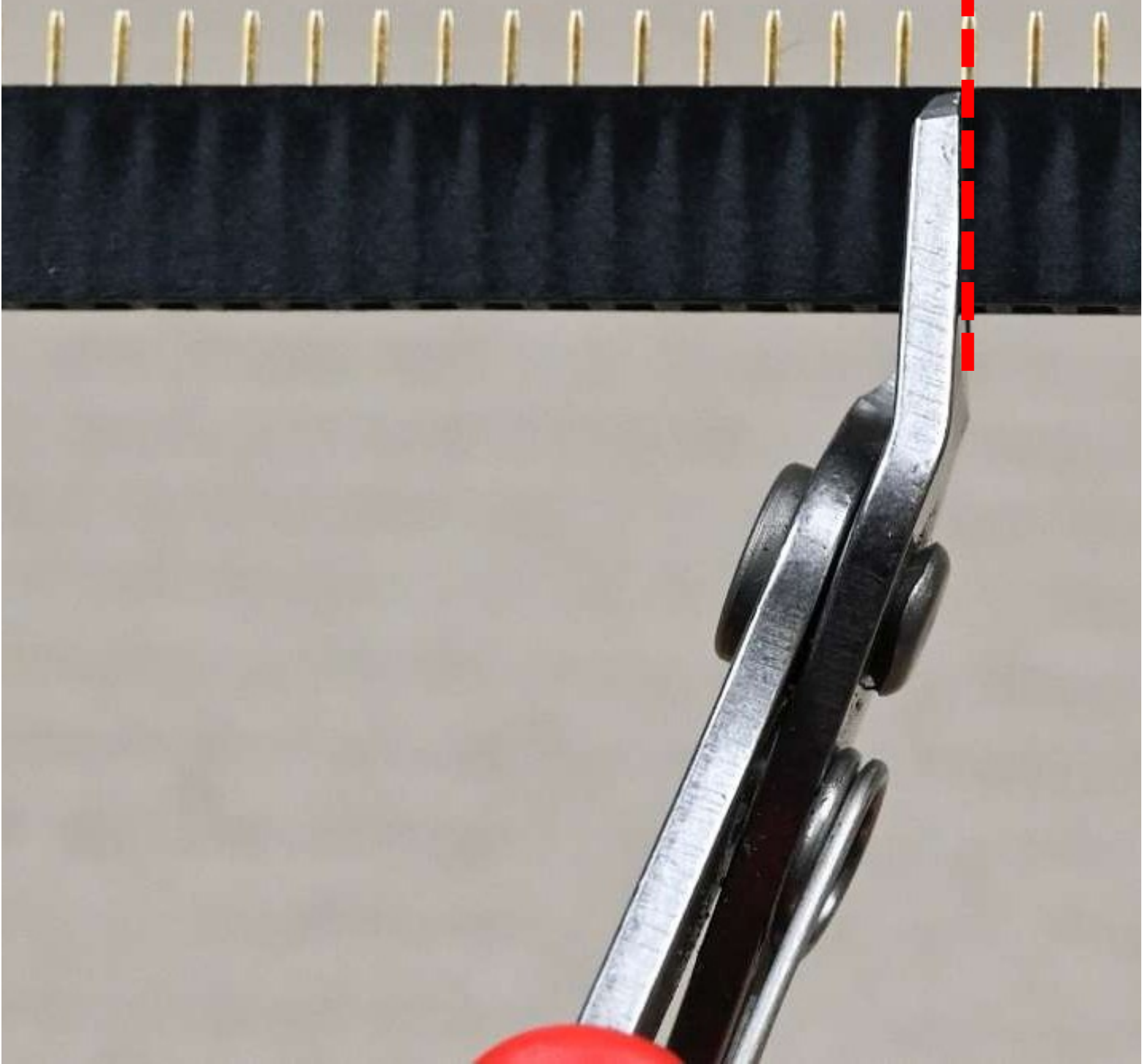
Analog 1

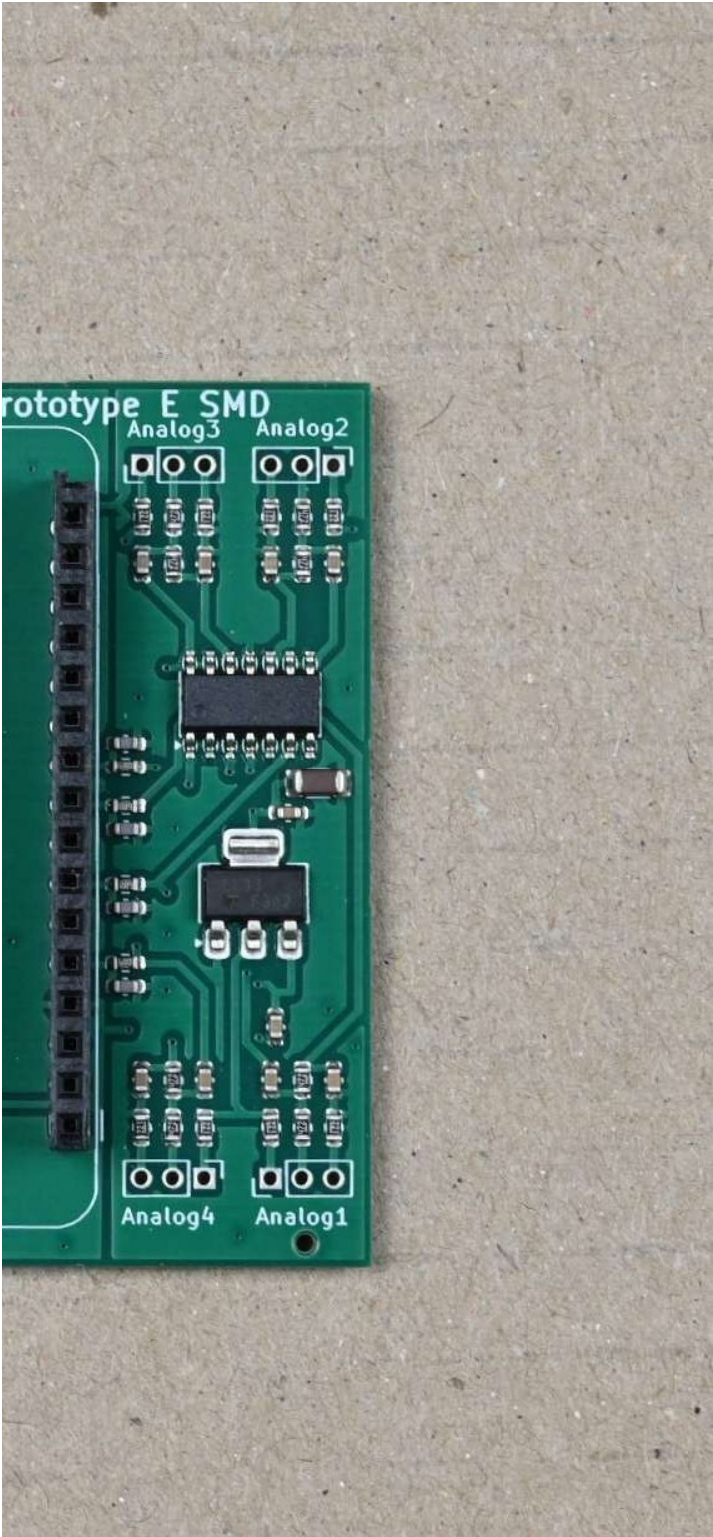
Analog 2

Switch

Cut here

16

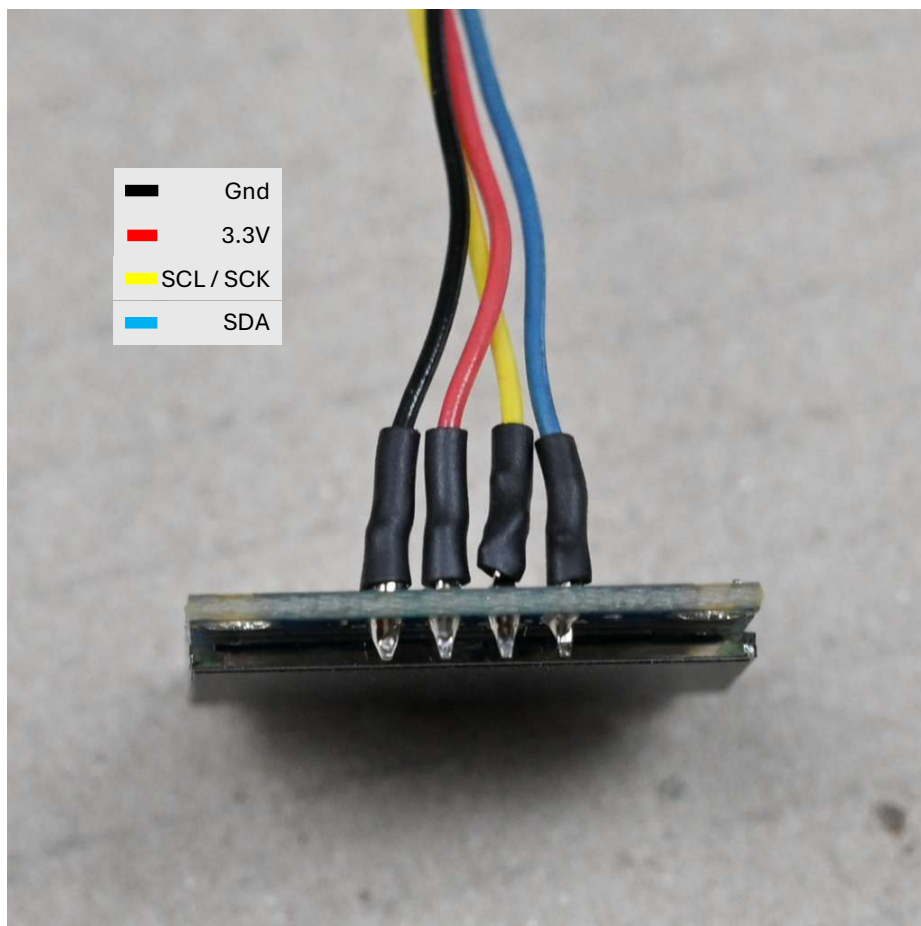


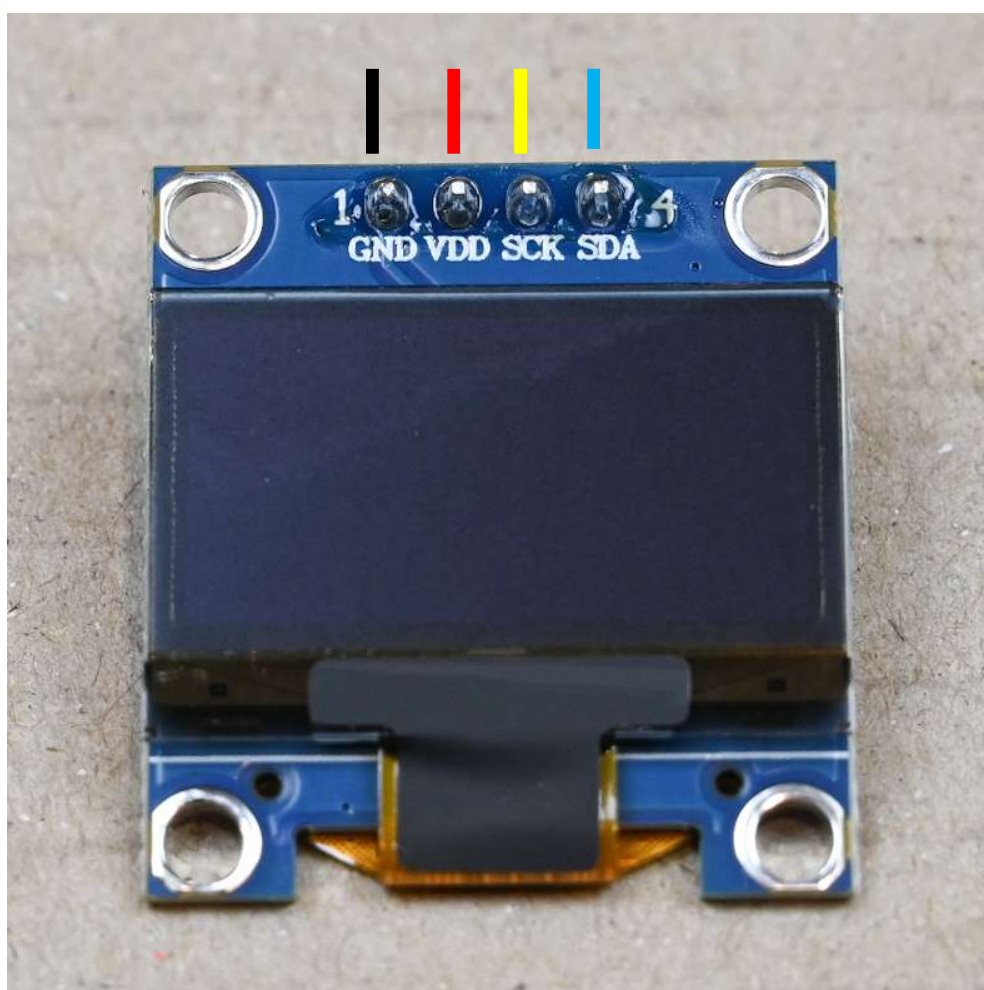


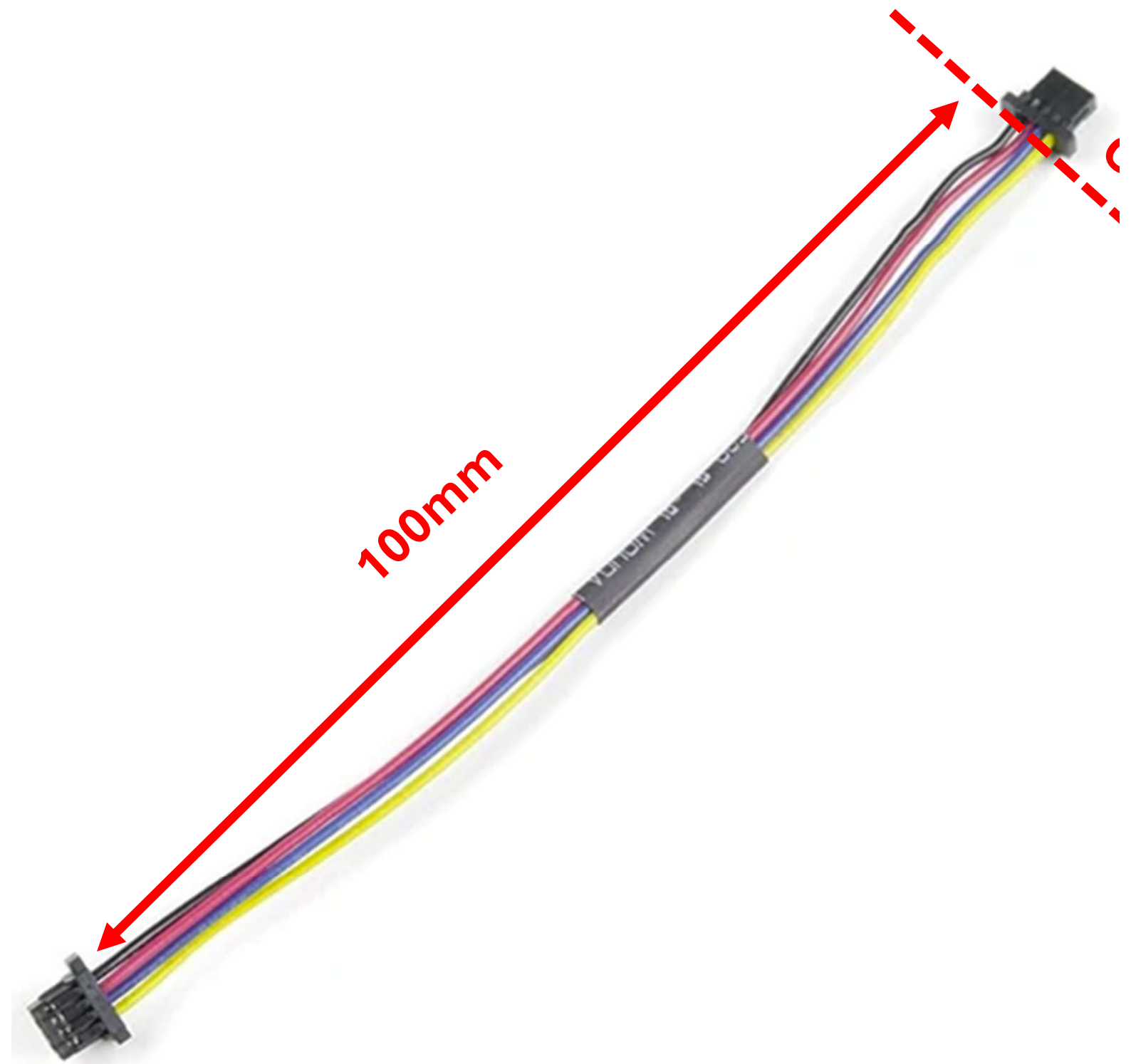
older











100mm

Measure length across insulation





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JLCCNC (0)

JLCMC (0)

Flex Heater (0)

Qty

Build Time

Price

shield Proto E SMD fab_Y9

type:Y9-3439461A

Thickness, HASL(with lead)

#: 

etails | Edit Order

5



2 days

A\$

A\$

shield Proto E SMD fab_Y9

PCBA: SMT026042460381-3...

top side

#: 

etails |

5

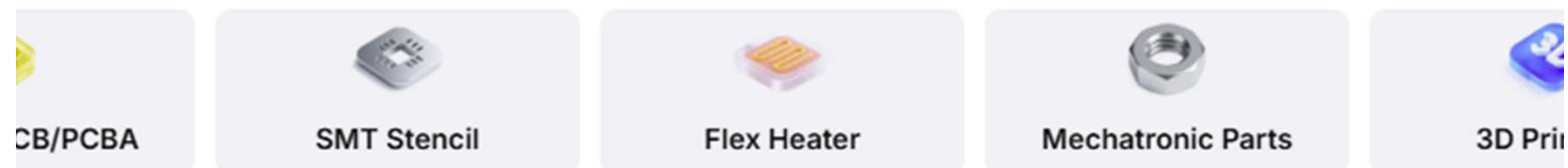


1 - 2 days

A\$



AUD ▼



[Instructions For Ordering>](#)

[Upload History ▼](#)

Charge Details

Special Offer

Via Covering

Surface Finish

PCB Build Time

☒ 2 days

☐ 24 hours

☐ 24 hours

Calculated Price

Additional charge

Add gerber file

Only accept zip or rar, Max 100 MB, [View example >](#)

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Flex



Aluminum



Copper Core



Rogers



PTFE Teflon

4

High Precision PCB

6



8

10

12

14

16

More ▼

100

mm ▼

[Consumer electronics](#)

Aerospace

Medical

Shipping Estimate

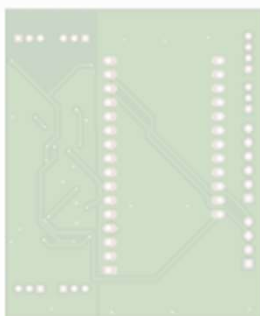
▼ DHL Express

Weight



COUPON

Assembly cost starting from \$0 with coupon



☐ Assemble bottom side

Standard

[What's the difference?](#)

One Side

Both Sides

Added by Customer

by JLCPCB

Charge Details

Special Offer

Via Covering

Surface Finish

Economic PCBA Price

Components

Extended components fee 

SMT Assembly 

Hand-soldering labor fee

Manual Assembly 

PCB Build Time

☒ 2 days

☐ 24 hours

☐ 24 hours **PCBA Only**

Calculated Price

Additional charges may apply for spe

NEXT

Shipping Estimate

 DHL Express

Weight 

Coupons 

Save A\$27.8

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placements

Quote & Order

Aut

o E SMD BOM.csv

[iew Sample BOM](#)



✓ BODAQs shield Prot

Pick&Place File, Only accept XLS,XL

Bottom

Top Designator	JLPCB Part #	Comment	Footprint
C5,C11,C15,C8...	C91183	100nF50...	0603
C6,C12,C3,C9	C76599	10nF5...	0603
C10,C7,C1,C4	C519554	47nF5...	0603
C17	C696375	10uF52...	1206
R6,R7,R16,R17...	C22962	-55?~+1...	0603
R10,R15,R20,R5	C27592	100mW...	0603
R14,R8,R13,R9...	C23162	-55?~+1...	0603
U2	C7378	MCP6004	SOIC-14...
U4	C435835	LDL1117...	SOT-223-4

NEXT

JLPCB

Top Side Total 9 parts detected 9 Parts confirmed

Uploaded BOM Data			
Top Designator	Comment	Footprint	Matched Part Detail
C5,C11,C15,C8,C21	100nF50VX7R 5%0603M...	0603	CC0603JRX7R9BB104 100nF 50V X7R ±5% 0603
C6,C12,C3,C9	10nF50VCO G5%060...	0603	C1608C0G1H103JT000N 10nF 50V C0G ±5% 0603
C10,C7,C1,C4	47nF50VX7 R5%060...	0603	CC0603JRX7R9BB473 47nF 50V X7R ±5% 0603
C17	10uF25V R5%120...	1206	C1206C106J3RACAUTO 10uF 25V X7R ±5% 1206
R6,R7,R16,R17,R12, R11,R1,R2	-55?~+155?10 0mW220?...	0603	0603WAF2200T5E -55°C~+155°C 100mW 220
R10,R15,R20,R5	100mW68? 75VThickF...	0603	0603WAF680JT5E -55°C~+155°C 100mW 68
R14,R8,R13,R9,R19, R4,R3,R18	-55?~+155?10 0mW4.7k?...	0603	0603WAF4701T5E -55°C~+155°C 100mW 4.7
U2	MCP6004	SOIC-14_3.9×8.7 mm_P1.27mm	MCP6004T-I/SL -40°C~+125°C 1.8V~6V 10
U4	LDL1117S33R	SOT-223-4	LDL1117S33R -40°C~+125°C@(Tj) 1.1.2A

+ Add Part

Coupons

APP Download

AUD

0

Order Now

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Automatically saved, last updated on 24 April, 12:24

Total Price:

A\$54.01

Weight

321.30g

Product Description

Select

PCBA Remark

SAVE TO CART

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SHOPPING CART

All (1)

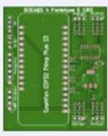
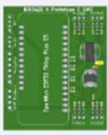
JLCPCB (1)

JLC3DP (0)

JLCCNC (0)

JLCMC (0)

Flex Heater (0)

	Item	Qty
☐	JLCPCB (PCB/PCBA/Stencil)	
☐	 BODAQS shield Proto E SMD fab_Y9 PCB prototype:Y9-3439461A Green, 1.6 Thickness, HASL(with lead) Customer #: Product Details Edit Order	5
☐	 BODAQS shield Proto E SMD fab_Y9 Economic PCBA: SMT026042460381-3... Assemble top side Customer #: Product Details	5

Recommended Deals For You

Refresh



Hex Socket Screw - Full/Partial Thread, Standard Type (DIN...

From \$0.0148

Order Now



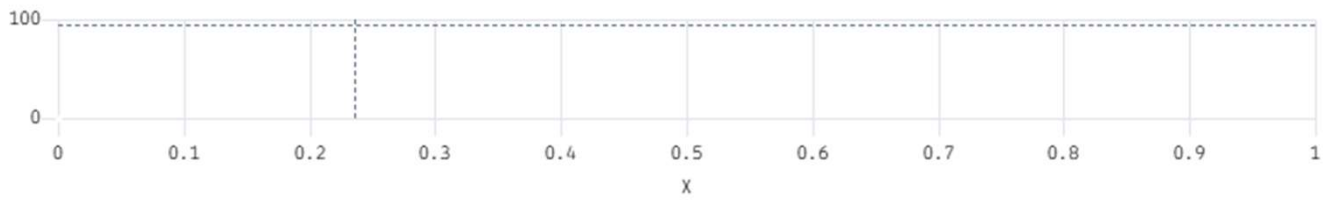
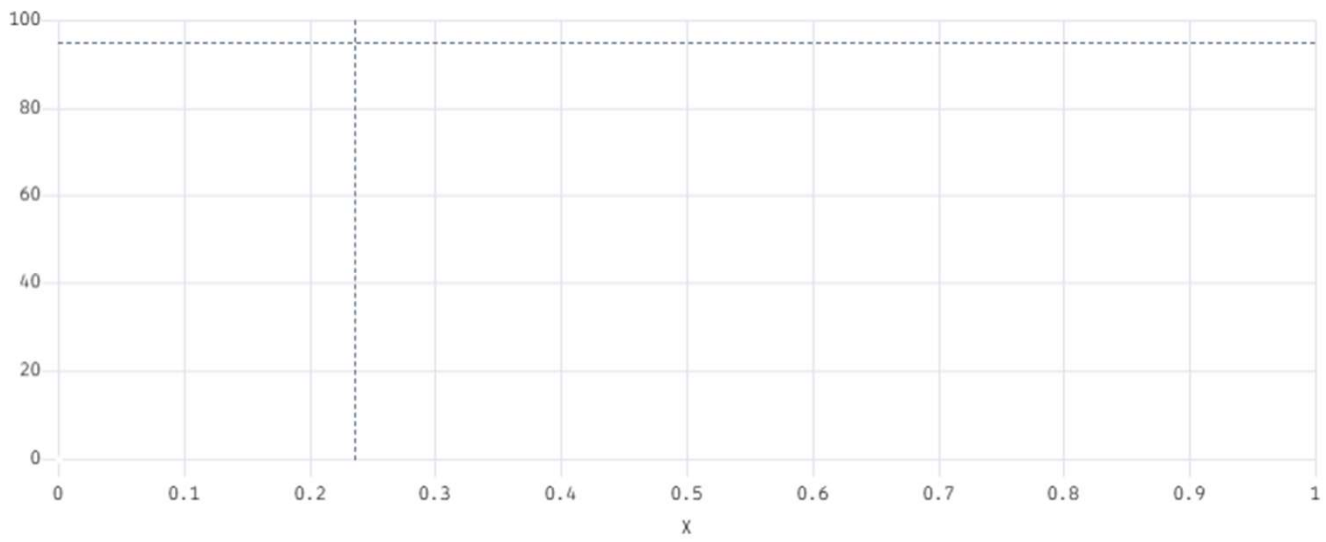
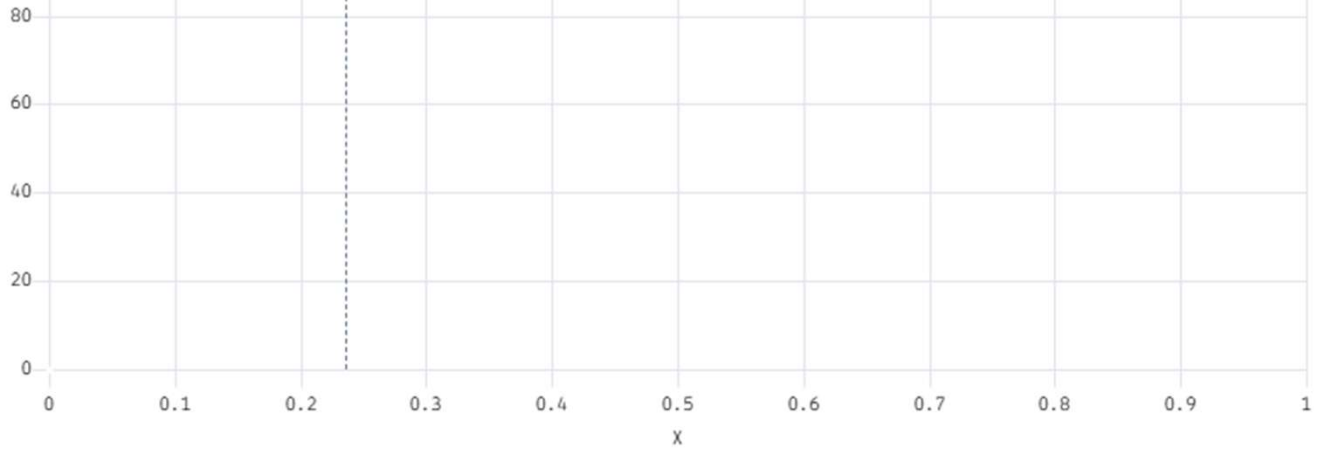
Hex Socket Screw - Full/Pa Thread, Standard Type (DI

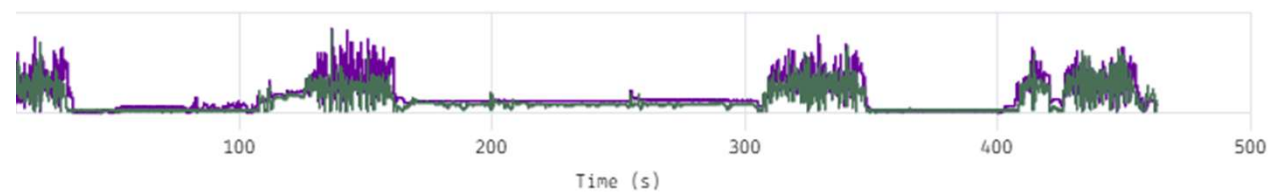
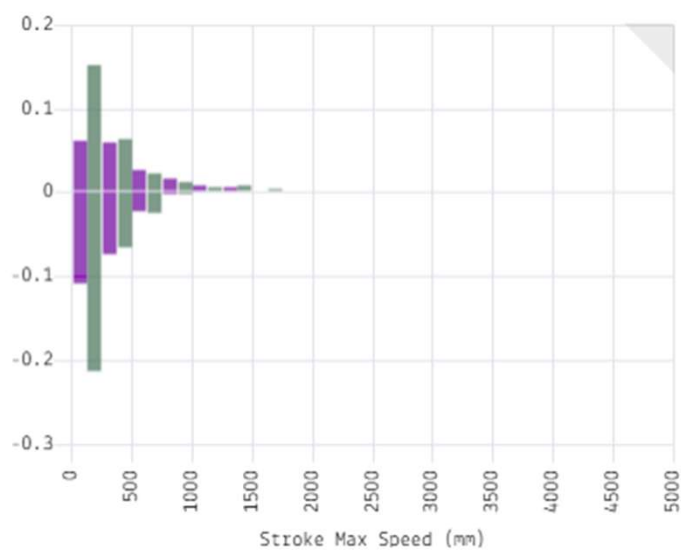
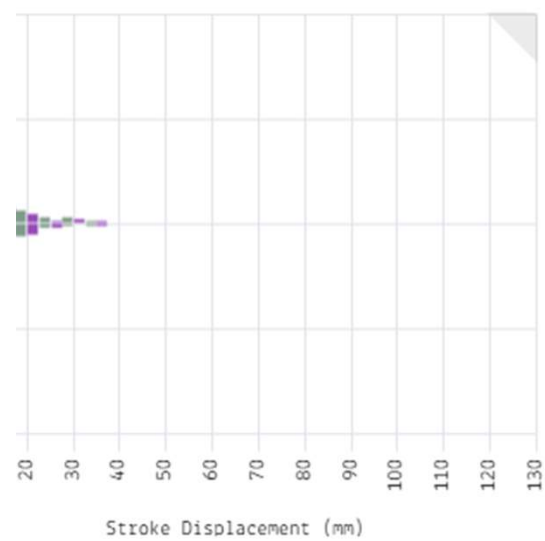
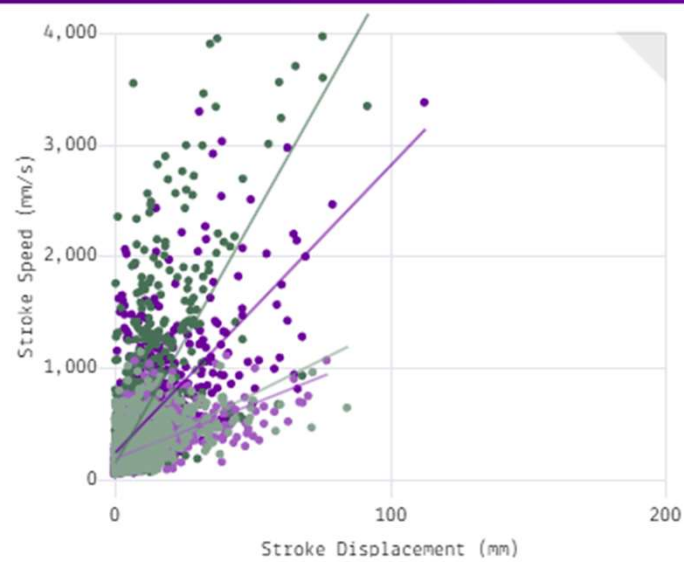
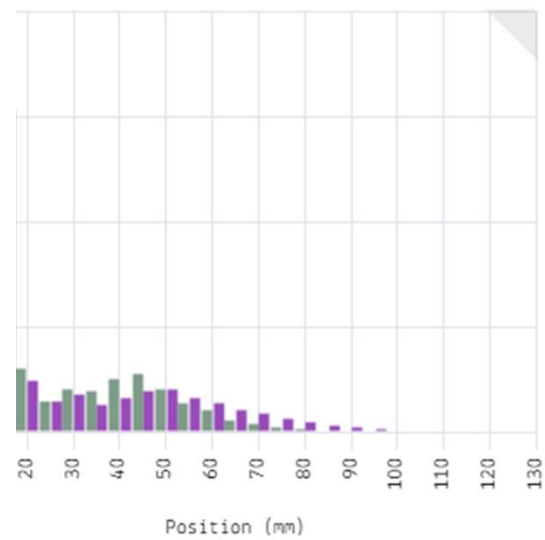
From \$0.0159

Order Now



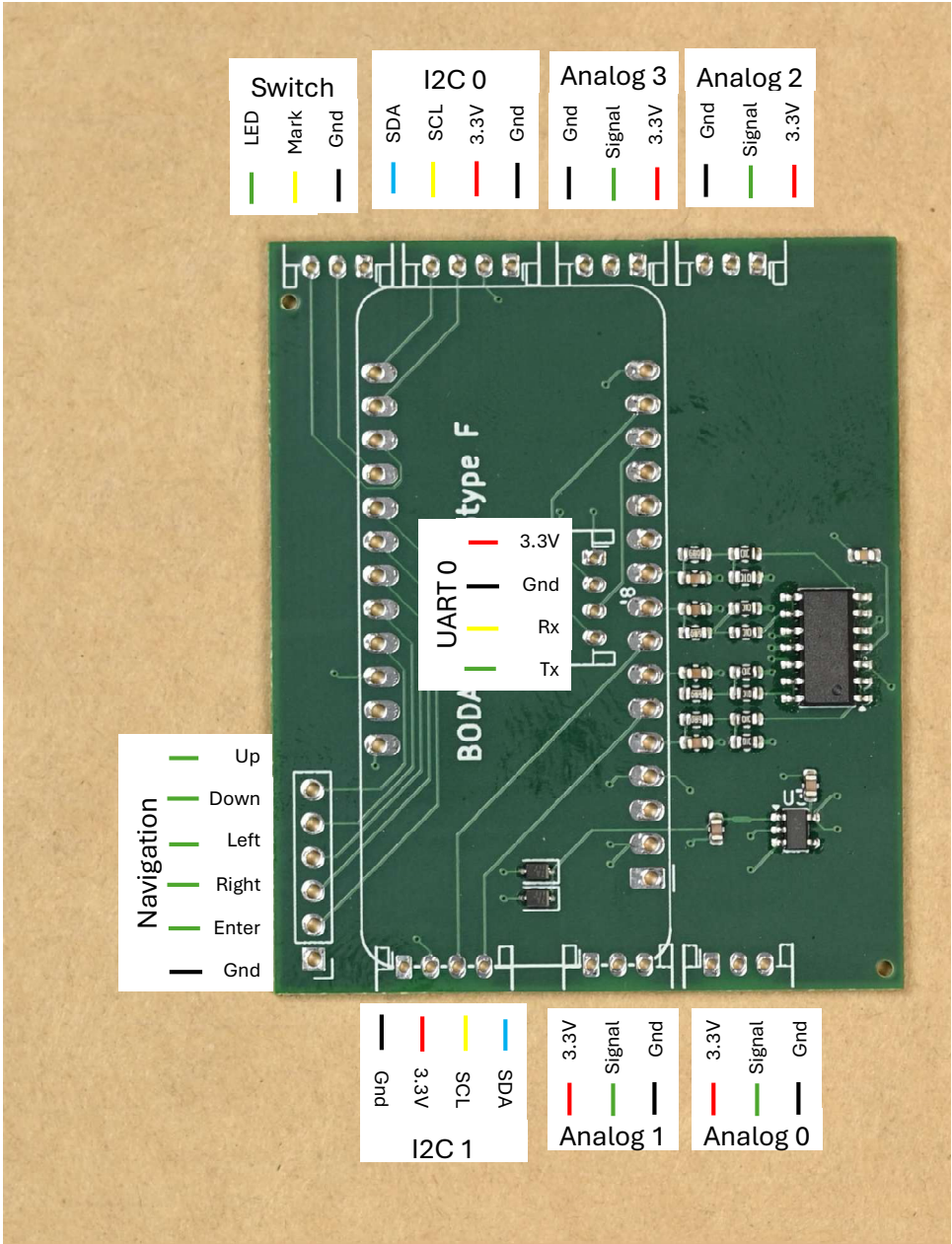
Server connected with ID: Xhod8KIkQVNuuRQEAAAB





Analog 2

3.3V
Signal
Gnd
SDA
SCL



Inside (solder side)

Switch

Analog 3

I2C 0

Analog 2

Analog 0

Analog 1

Outside (connector side)

Analog 3

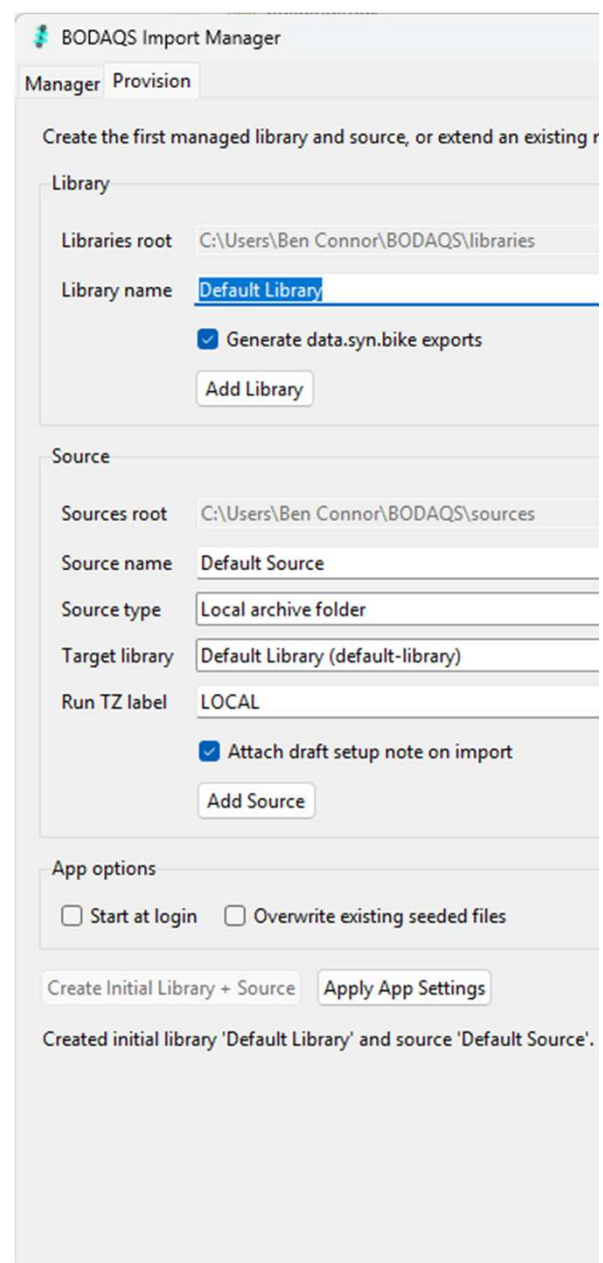
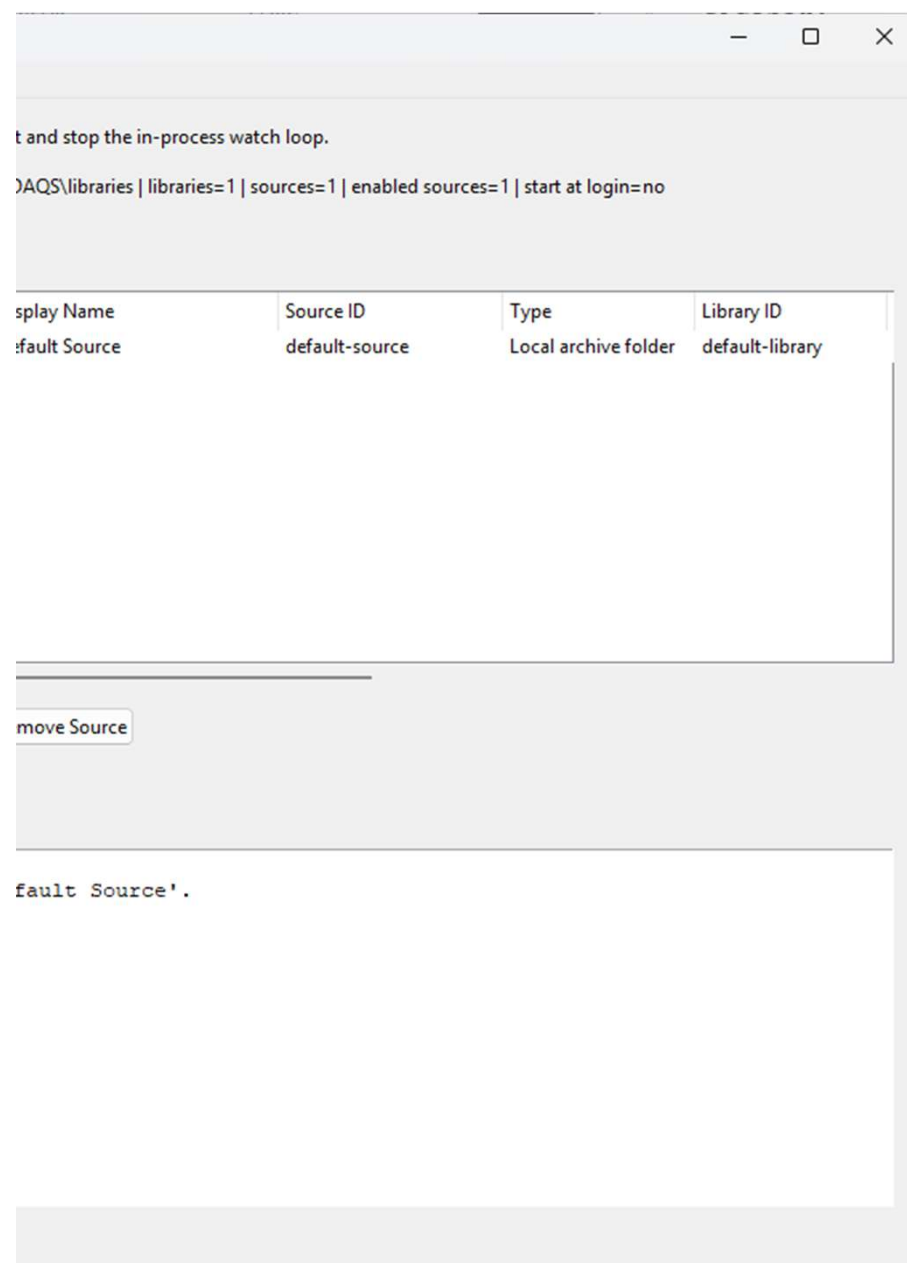
Switch

Analog 1

Analog 0

Analog 2

I2C 0



or extend an existing managed setup with another library or source.

5\libraries

e exports

5\sources

on import Session base name

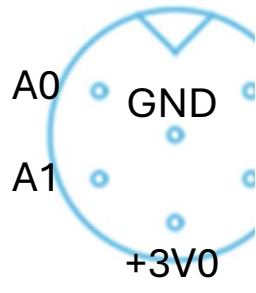
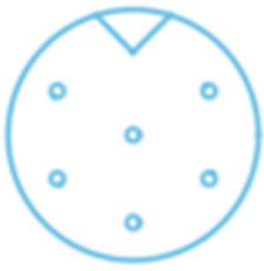
te on import

seeded files

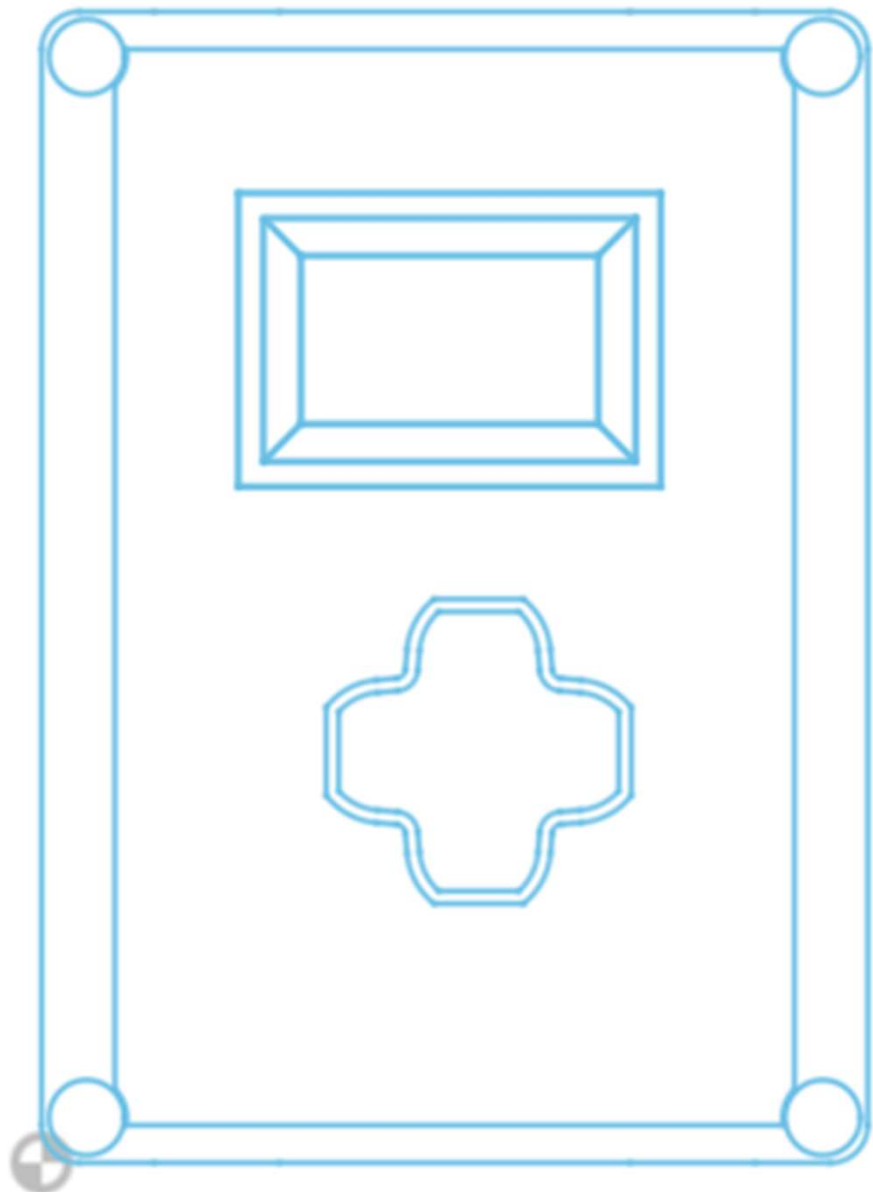
ig Workspace

Apply App Settings

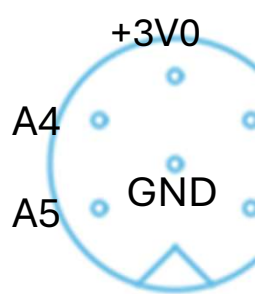
setup.

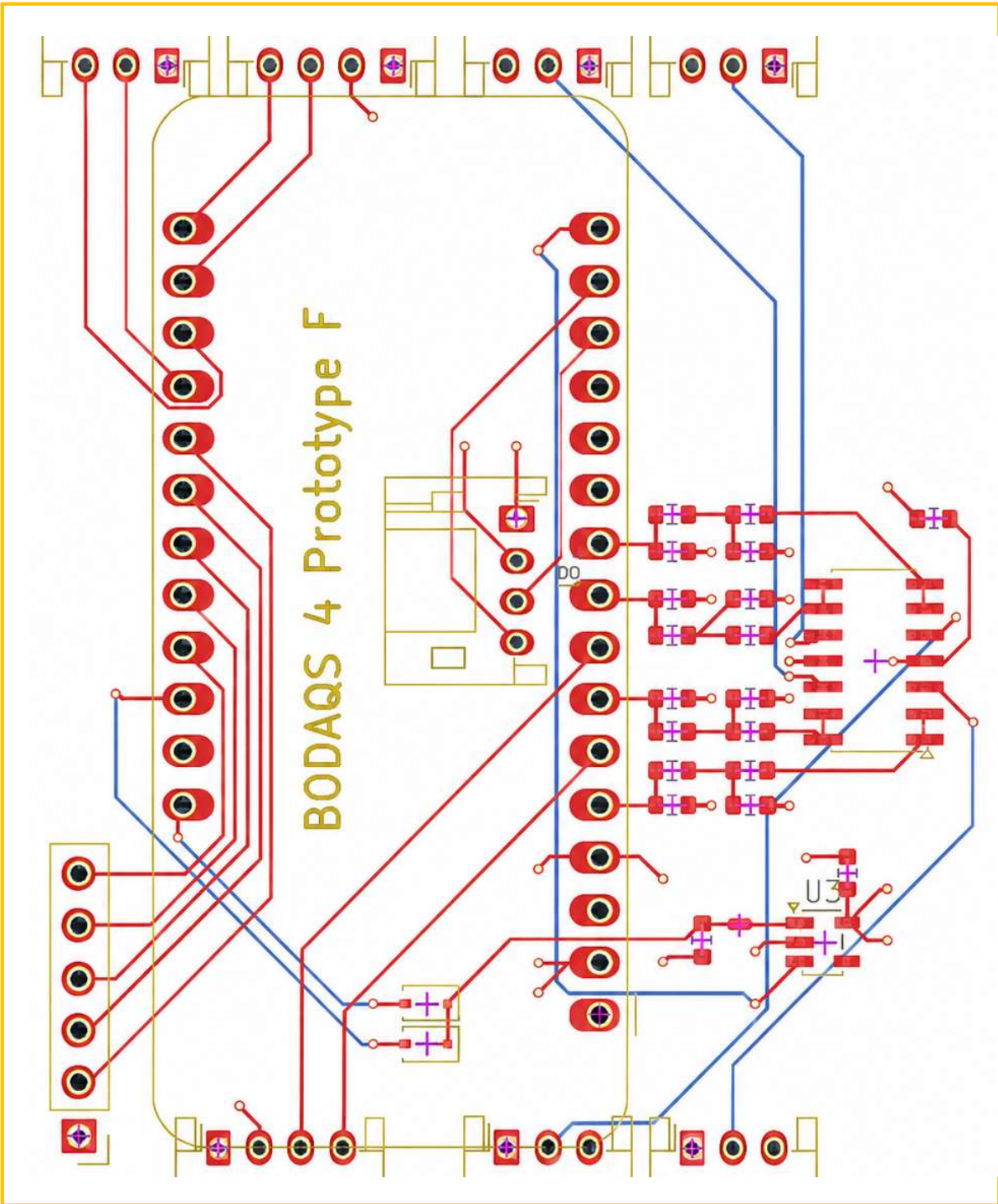


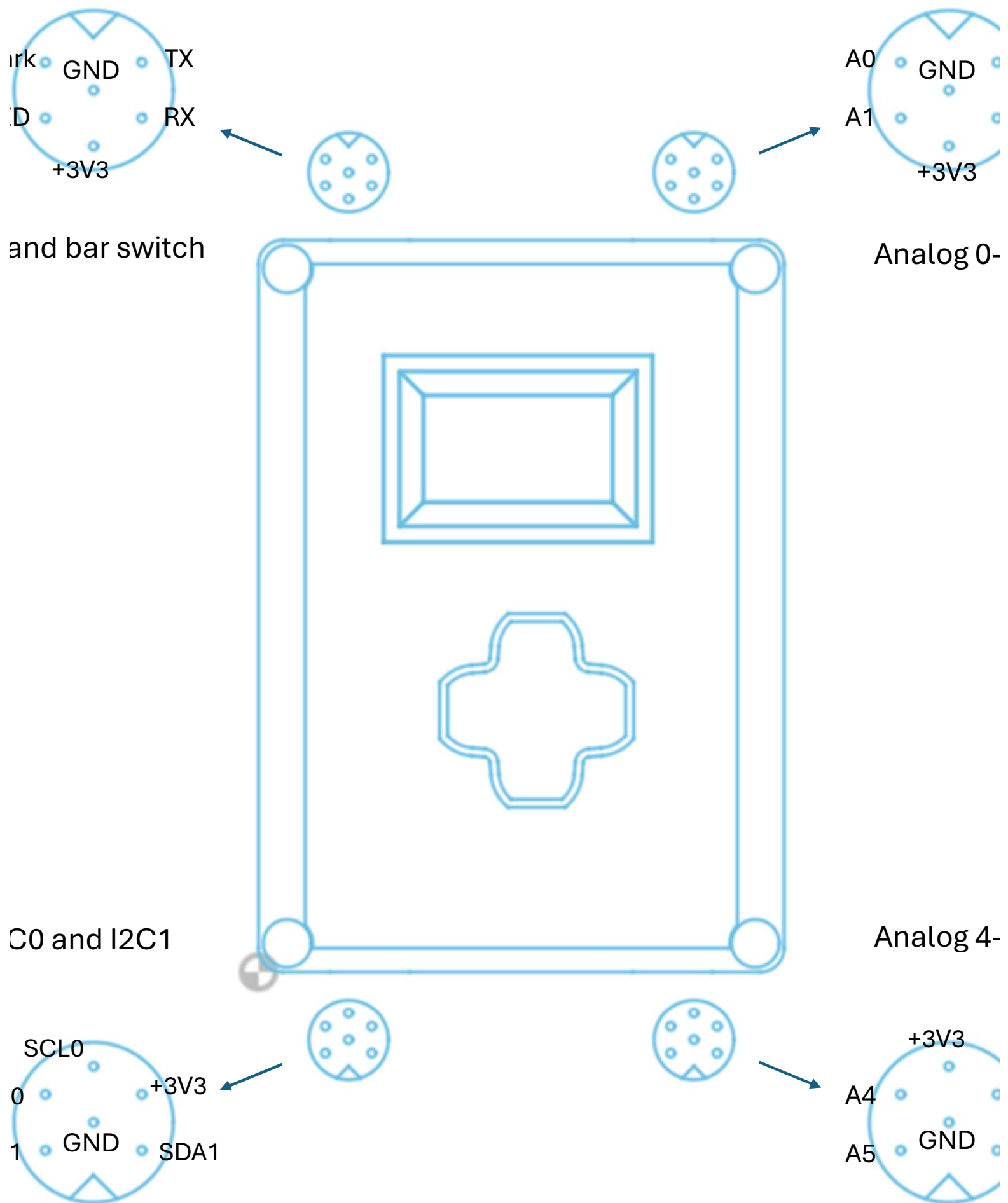
Analog 0-

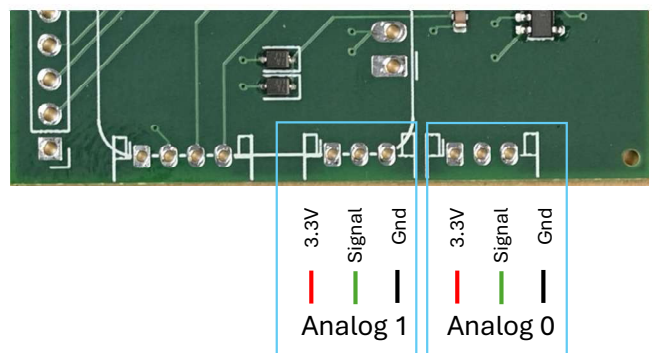
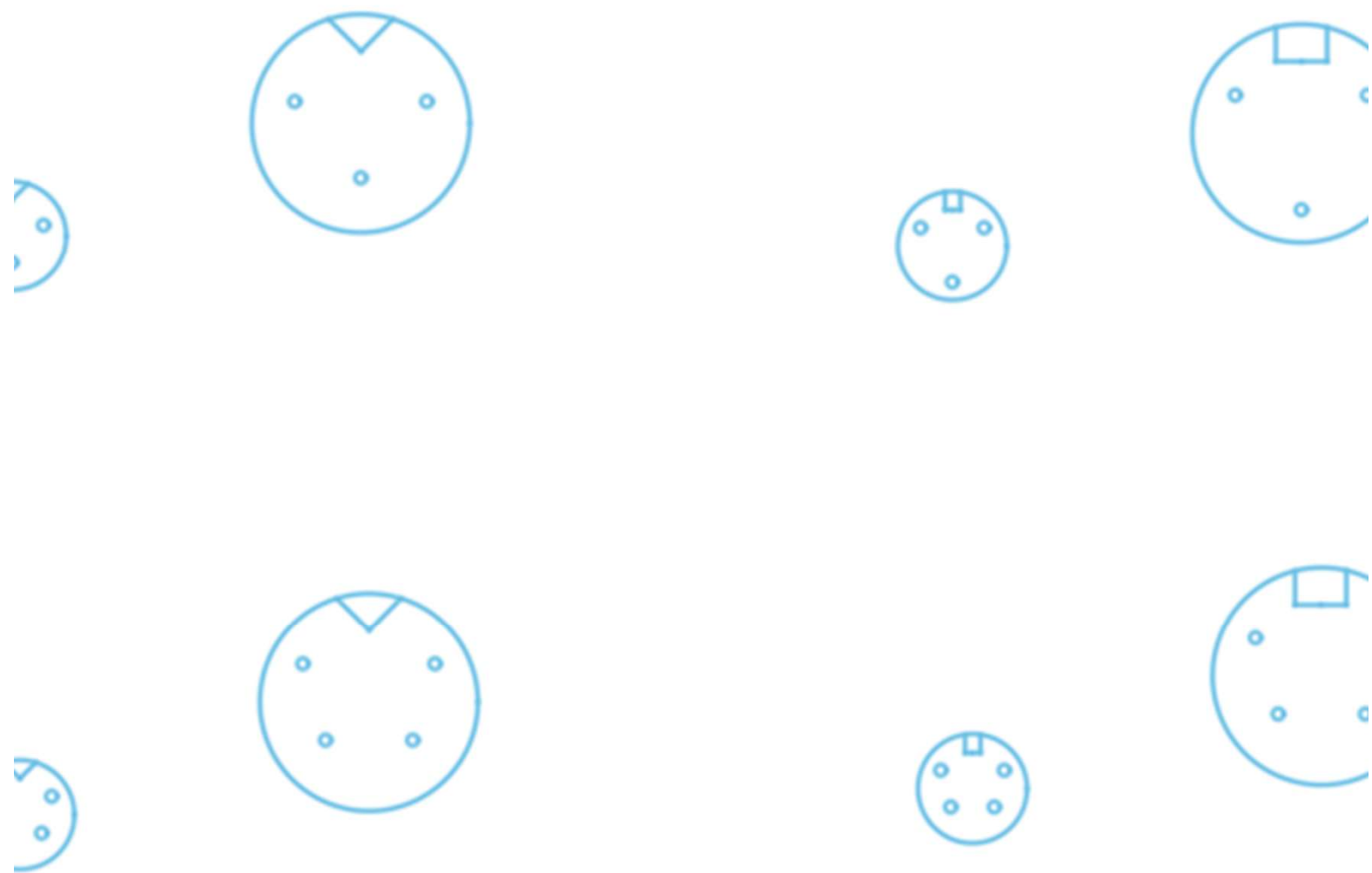


Analog 4-



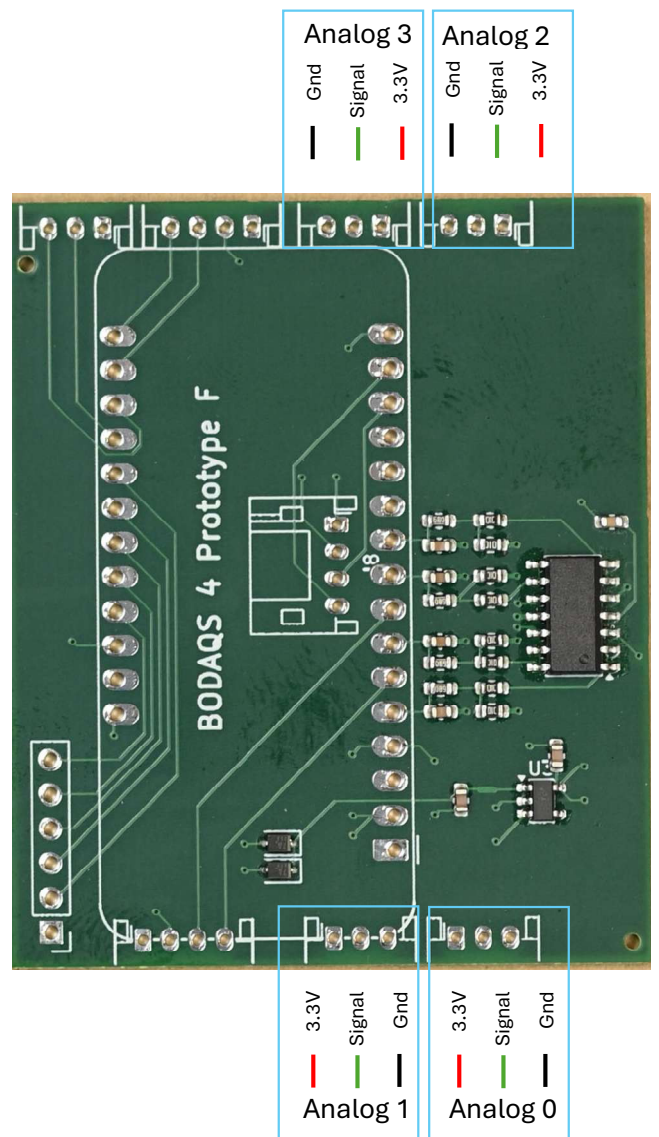


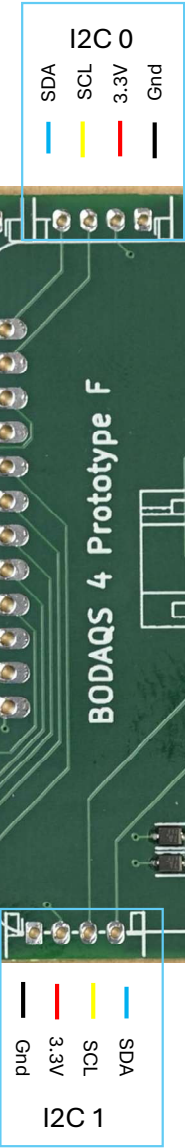
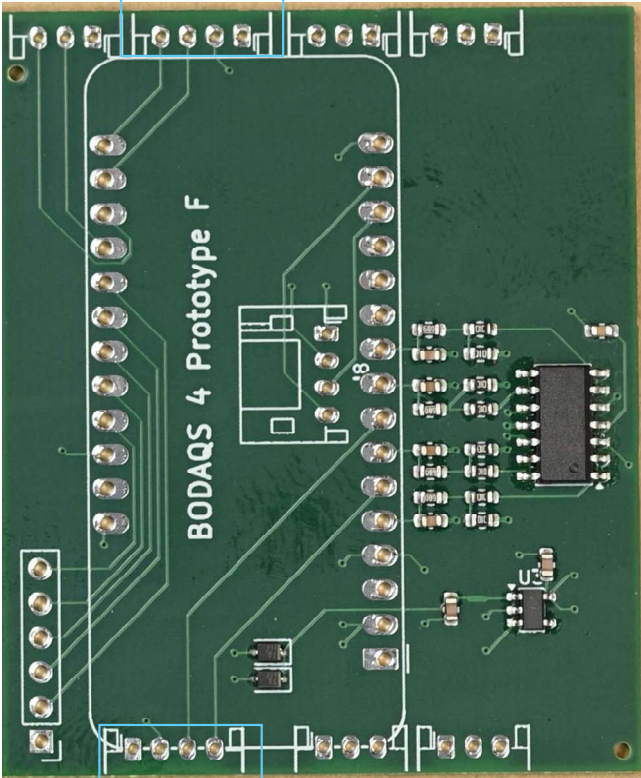


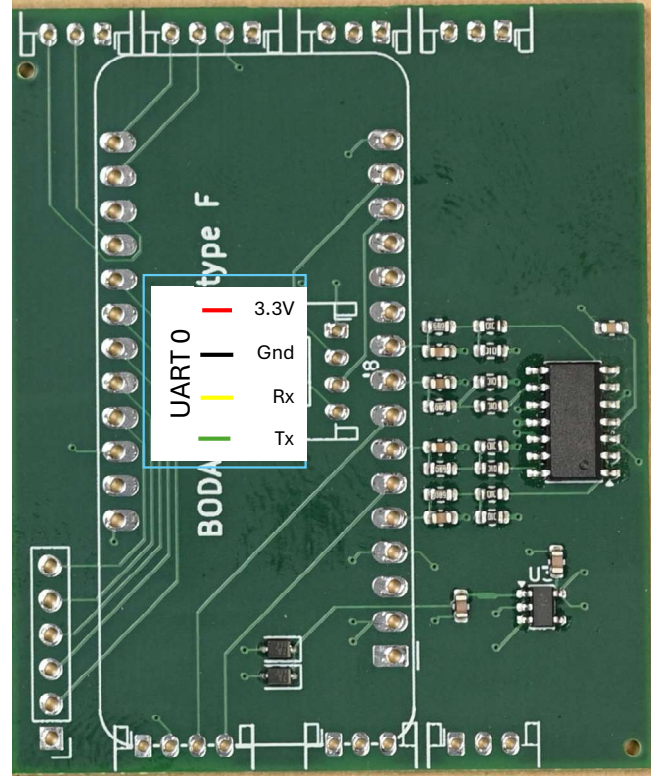


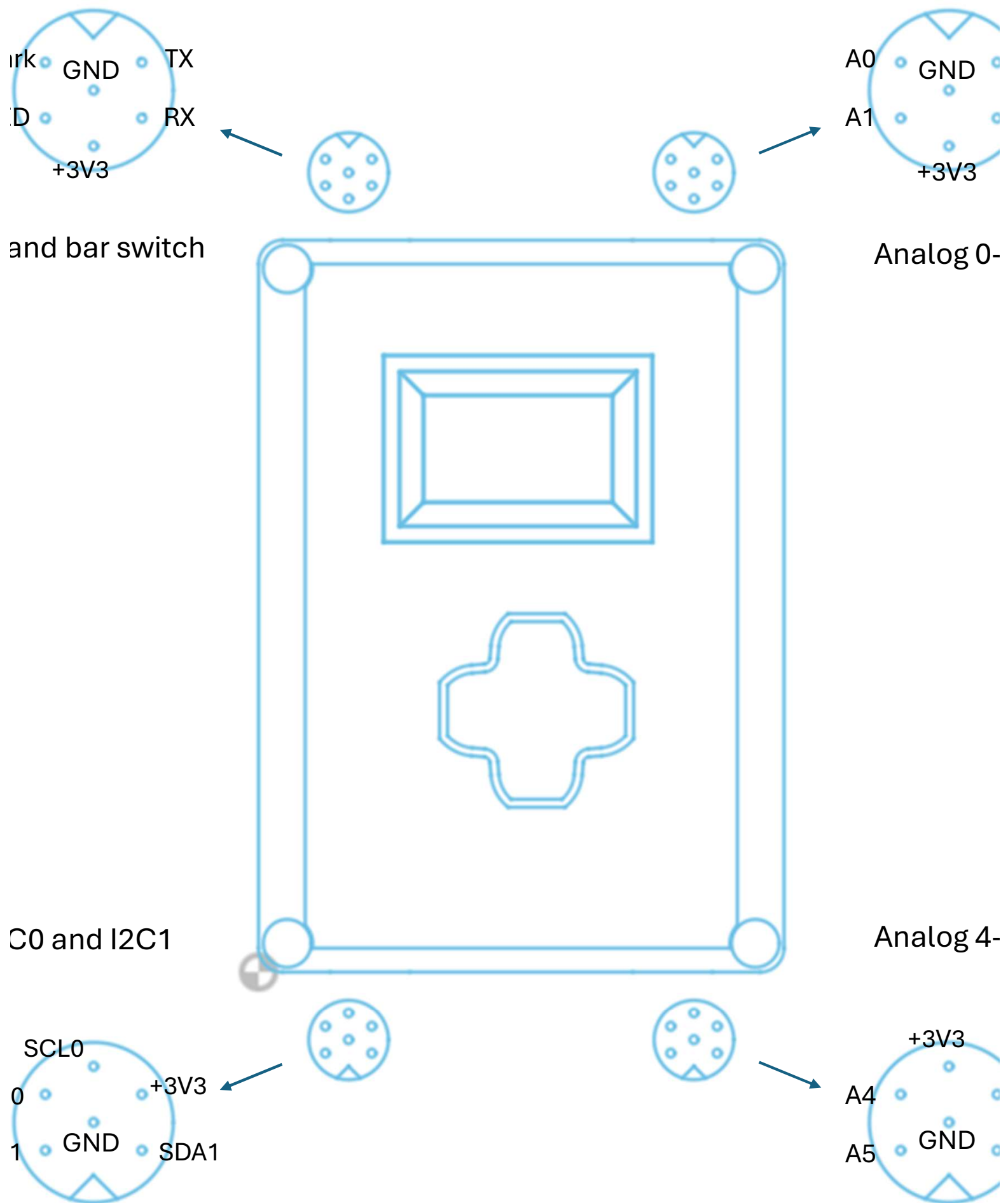
Analog 2

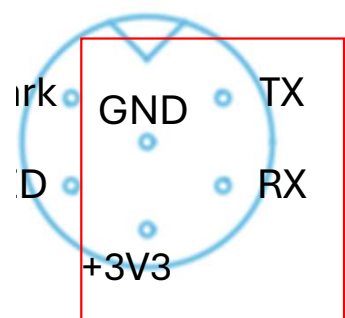
3.3V
Signal
Gnd
SDA
SCL



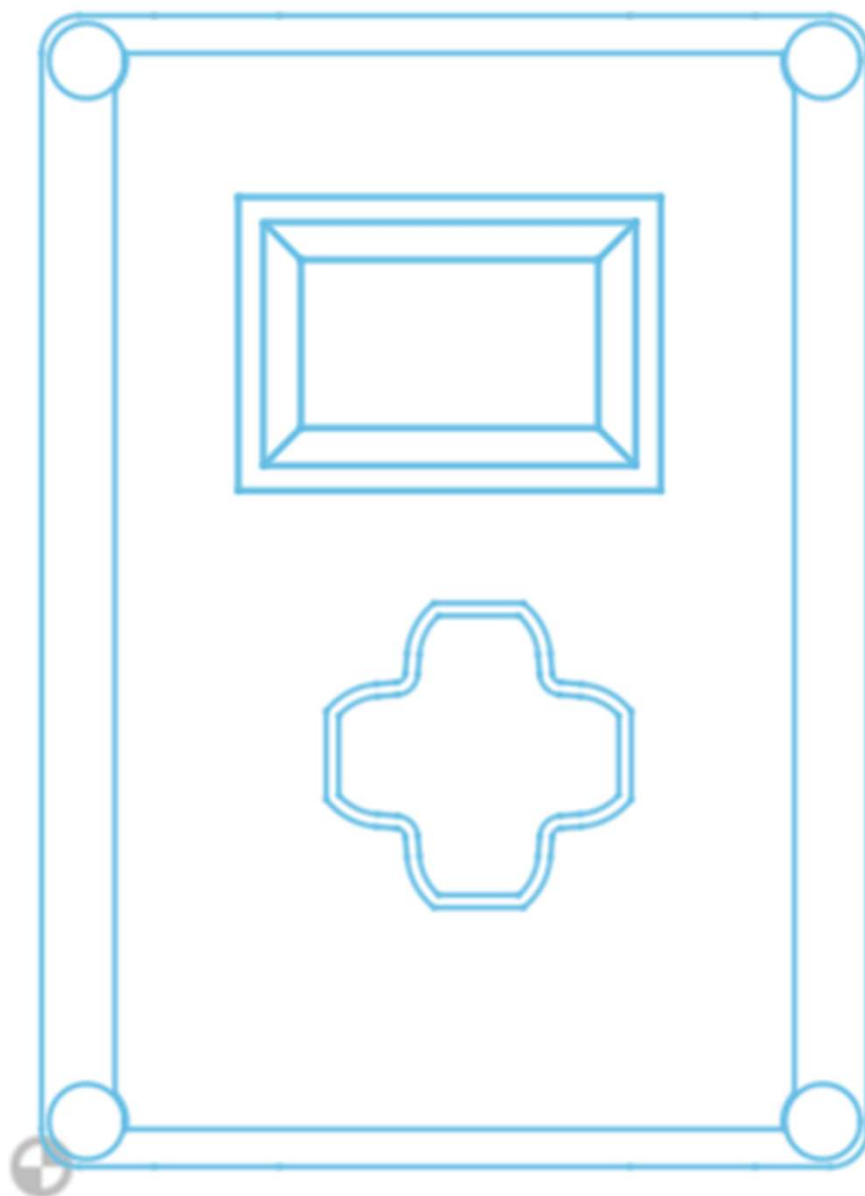


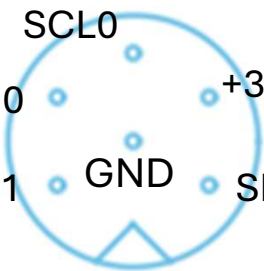
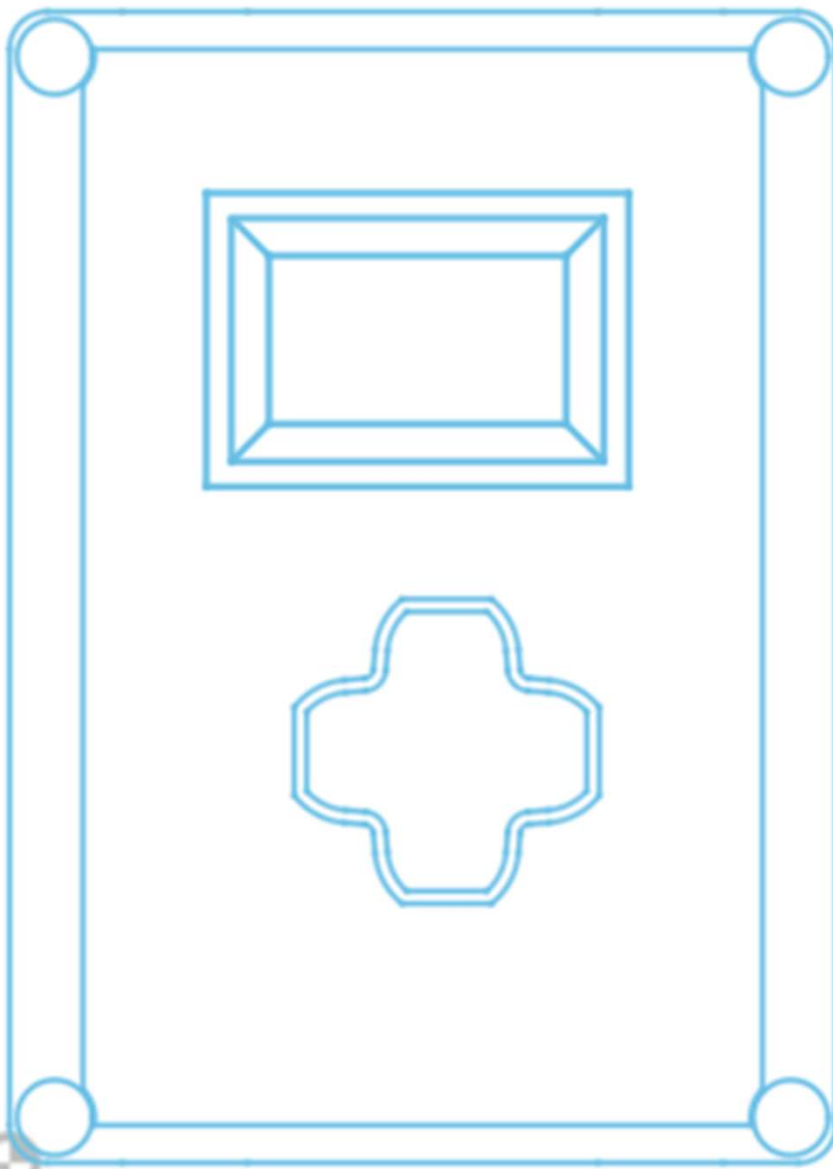
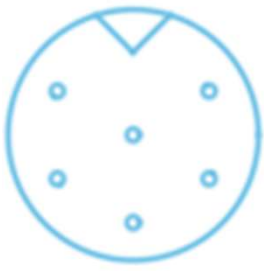






and bar switch

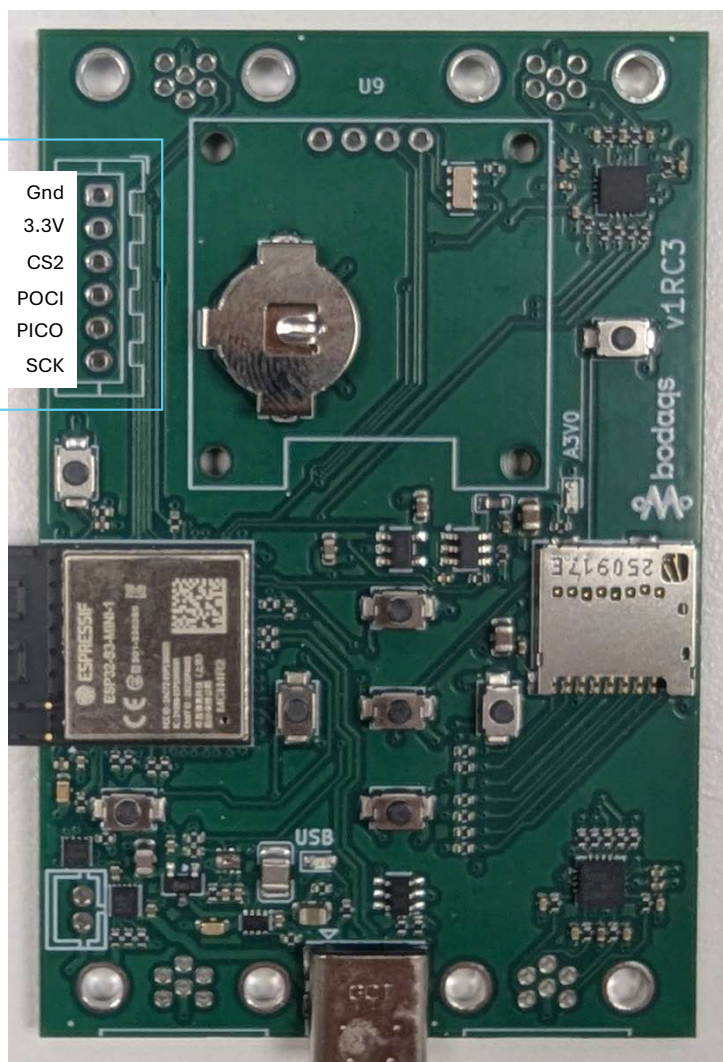
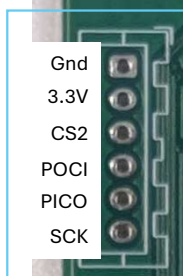




C0 and I2C1

SCL0
0
1
+3V3
GND
SDA1

UART 0





aqs_rc3_001

▼

Leave blank to auto-name.

aqs_rc3_001

▼

Leave blank to auto-name.

JS data logger: bodaqs_rc3_001

ss point / bodaqs_rc3_001 IP: 192.168.4.1

[Sensors](#)

5

ck - Analog Potentiometer

front_shock

Analog Potentiometer

Apply Type

Rebuilds fields for this sensor type; restart after add/delete/type changes.

ut

AIN0 (ADS0 CH0)

1

efault

☐

de

LINEAR

Complex transforms are applied downstream during import/analysis.

2

y column

☒ Append RAW column after primary

travel (mm)

203

If 1 => normalized; > 1 => mm

3

nge

170

Installed range in linear output units for sag percentage

4

front

Optional semantic end for log metadata

main

suspension

Optional semantic domain for primary output

antity

disp

Optional semantic quantity for primary output

5

methods

ZERO,RANGE

Read-only sensor capability. Calibration values below remain editable.

nt at zero travel

0

Counts at sensor 0 position

nt at full travel

4095

Counts at sensor full scale position

ro count

0

Installed zero point (counts)

6

Delete this sensor

Removes the sensor from logger config only.

ata logger: bodaqs_rc3_001

/ bodaqs_rc3_001 IP: 192.168.4.1

[DRS](#)

55600 String Pot (Analog)

AS5600 String Pot (Analog) ▼

Apply Type

Rebuilds fields for this sensor type; restart after add/delete/type changes.

AIN1 (ADS0 CH1) ▼

1

☐

LINEAR ▼

Complex transforms are applied downstream during import/analysis.

2

☒ Append wrapped and unwrapped RAW columns

nm)

Full sensor travel in mm for RANGE scaling

3

Installed range in linear output units for sag percentage

4

front ▼

Optional semantic end for log metadata

suspension ▼

Optional semantic domain for primary output

disp ▼

Optional semantic quantity for primary output

5

is

Read-only sensor capability. Calibration values below remain editable.

o travel

Unwrapped counts at zero travel

l travel

Unwrapped counts at full travel

t

Installed zero point in unwrapped counts

6

Wrapped counts per AS5600 turn

unts)

Delta threshold used to detect wrap crossings

g start ☒

Reset unwrap state to turn 0 at each logging start

Delete this sensor

Removes the sensor from logger config only.

7

data logger: bodaqs_rc3_001

it / bodaqs_rc3_001 IP: 192.168.4.1

[sors](#)

GPS (UART)

GPS

DAN-F10N GPS (UART)

▼

Apply Type

Rebuilds fields for this sensor type; restart after add/delete/type changes.

☐

ds

NONE

Read-only sensor capability. Calibration values below remain editable.

0

▼

Board UART port index

38400

UART baud rate; DAN-F10N default is 38400

1

▼

GNSS navigation update rate

1500

Snapshot age after which position/motion values become NaN; 0 disables stale filtering

valid_only

▼

Controls when position/motion values are emitted

normal

▼

Temporary GPS isolation mode for web/GNSS interaction debugging

minimal

▼

GPS status columns to append

☒

Emit latitude and longitude columns

☒

Emit altitude column

☒

Emit speed and heading columns

4096

UART RX buffer size

1000

GNSS begin wait per background retry

500

GNSS configuration wait

Delete this sensor

Removes the sensor from logger config only.

1

2

3

4

AS5600 String Pot (I2C)

string_I2C

AS5600 String Pot (I2C)

Apply Type

Rebuilds fields for this sensor type; restart after add/delete/type changes.

1 I2C bus index

54

I2C address in decimal (default 54 = 0x36)

☐

LINEAR

Complex transforms are applied downstream during import/analysis.

☒ Append wrapped and unwrapped RAW columns

250

Full sensor travel in mm for RANGE scaling

170

Installed range in linear output units for sag percentage

front

Optional semantic end for log metadata

suspension

Optional semantic domain for primary output

disp

Optional semantic quantity for primary output

ZERO,RANGE

Read-only sensor capability. Calibration values below remain editable.

0

Unwrapped counts at zero travel

4095

Unwrapped counts at full travel

0

Installed zero point in unwrapped counts

4096

Wrapped counts per AS5600 turn

2048

Delta threshold used to detect wrap crossings

☒ Reset unwrap state to turn 0 at each logging start

0

Async I2C acquisition rate; 0 follows logger sample rate

Delete this sensor

Removes the sensor from logger config only.

ngle - AS5600 Angle (I2C)

5600_angle

AS5600 Angle (I2C) ▼ Apply Type Rebuilds fields for this sensor type; restart after add/delete/type changes.

1 ▼ I2C bus index

54 I2C address in decimal (default 54 = 0x36)

☐

mode

LINEAR ▼ Complex transforms are applied downstream during import/analysis.

raw column

☒ Append raw absolute angle counts after primary

l range

29.1 Installed range in linear output units for sag percentage

domain

rear ▼ Optional semantic end for log metadata

suspension ▼ Optional semantic domain for primary output

ang_disp ▼ Optional semantic quantity for primary output

ion methods

ZERO Read-only sensor capability. Calibration values below remain editable.

0 Firmware zero point in raw AS5600 counts

_mode

repeated ▼ I2C register read transaction style

counts_increase_positive ▼ Raw-count direction for positive angular movement

2x ▼ Volatile AS5600 slow filter setting; not burned to OTP

0 Async I2C acquisition rate; 0 follows logger sample rate

☐ Append AS5600 ANGLE register counts for diagnostics

☐ Append AS5600 AGC, status, magnitude, read-state, and failure-counter columns

250 Minimum interval between AS5600 diagnostic reads

Delete this sensor Removes the sensor from logger config only.

S5048B Angle (I2C)

5048_angle

AS5048B Angle (I2C) ▼ Apply Type Rebuilds fields for this sensor type; restart after add/delete/type changes.

0 ▼ I2C bus index

64 ▼ I2C address in decimal (default 64 = 0x40)

☐

1

LINEAR ▼ Complex transforms are applied downstream during import/analysis.

☒ Append raw absolute angle counts after primary

59.8 ▼ Installed range in linear output units for sag percentage

2

3

rear ▼ Optional semantic end for log metadata

suspension ▼ Optional semantic domain for primary output

ang_disp ▼ Optional semantic quantity for primary output

4

ZERO ▼ Read-only sensor capability. Calibration values below remain editable.

0 ▼ Firmware zero point in raw AS5048B counts

5

repeated ▼ I2C register read transaction style; repeated is the AS5048B datasheet path

counts_increase_positive ▼ Raw-count direction for positive angular movement

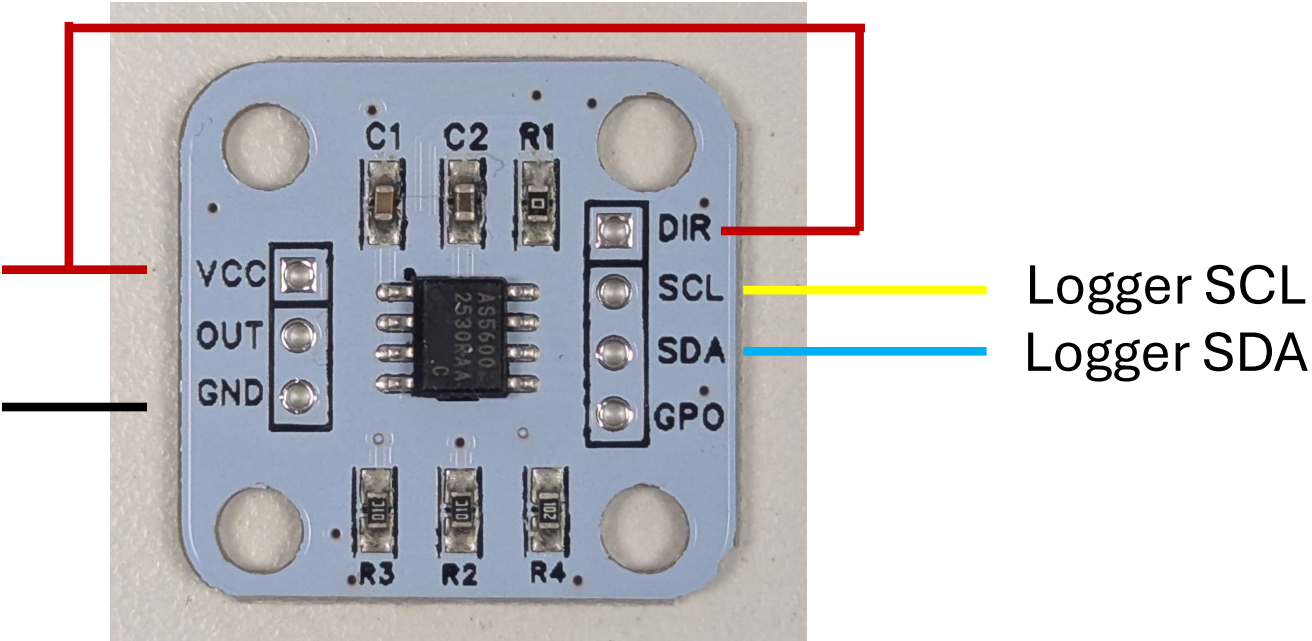
0 ▼ Async I2C acquisition rate; 0 follows logger sample rate

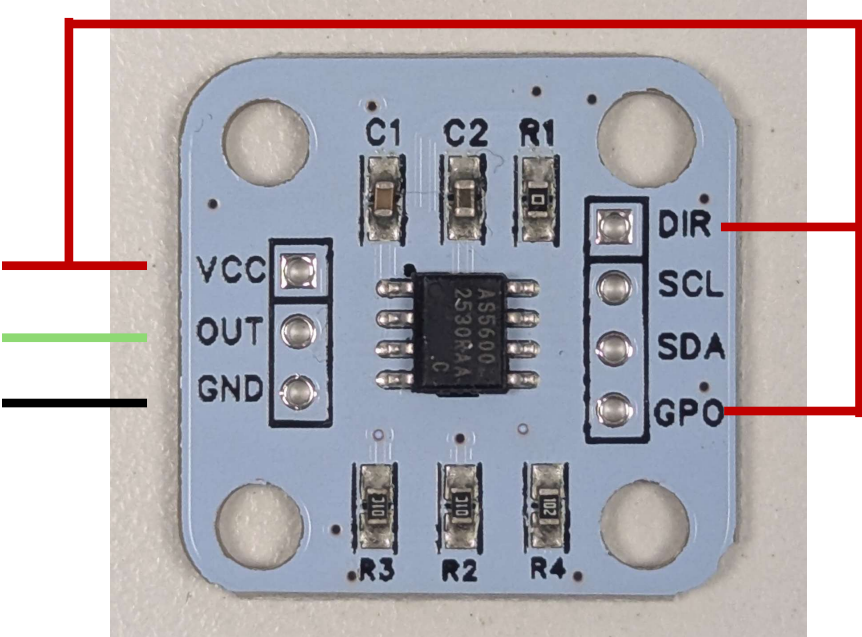
☐ Append AS5048B AGC, diagnostic flags, and magnitude columns

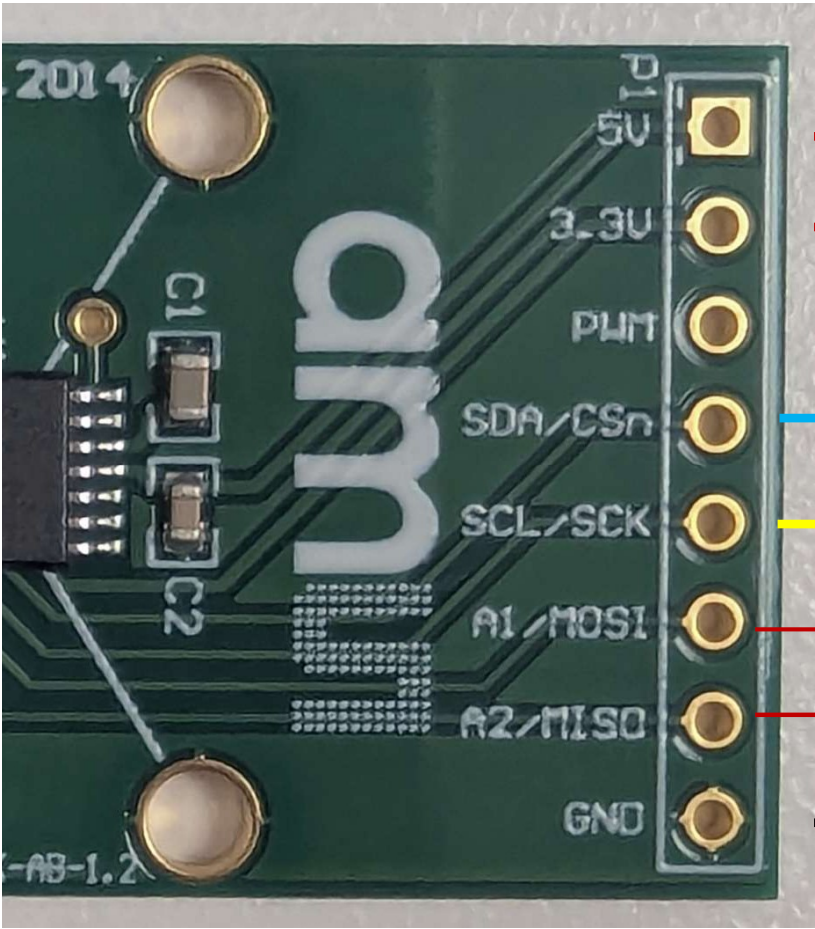
250 ▼ Minimum interval between AS5048B diagnostic reads

Delete this sensor Removes the sensor from logger config only.

6





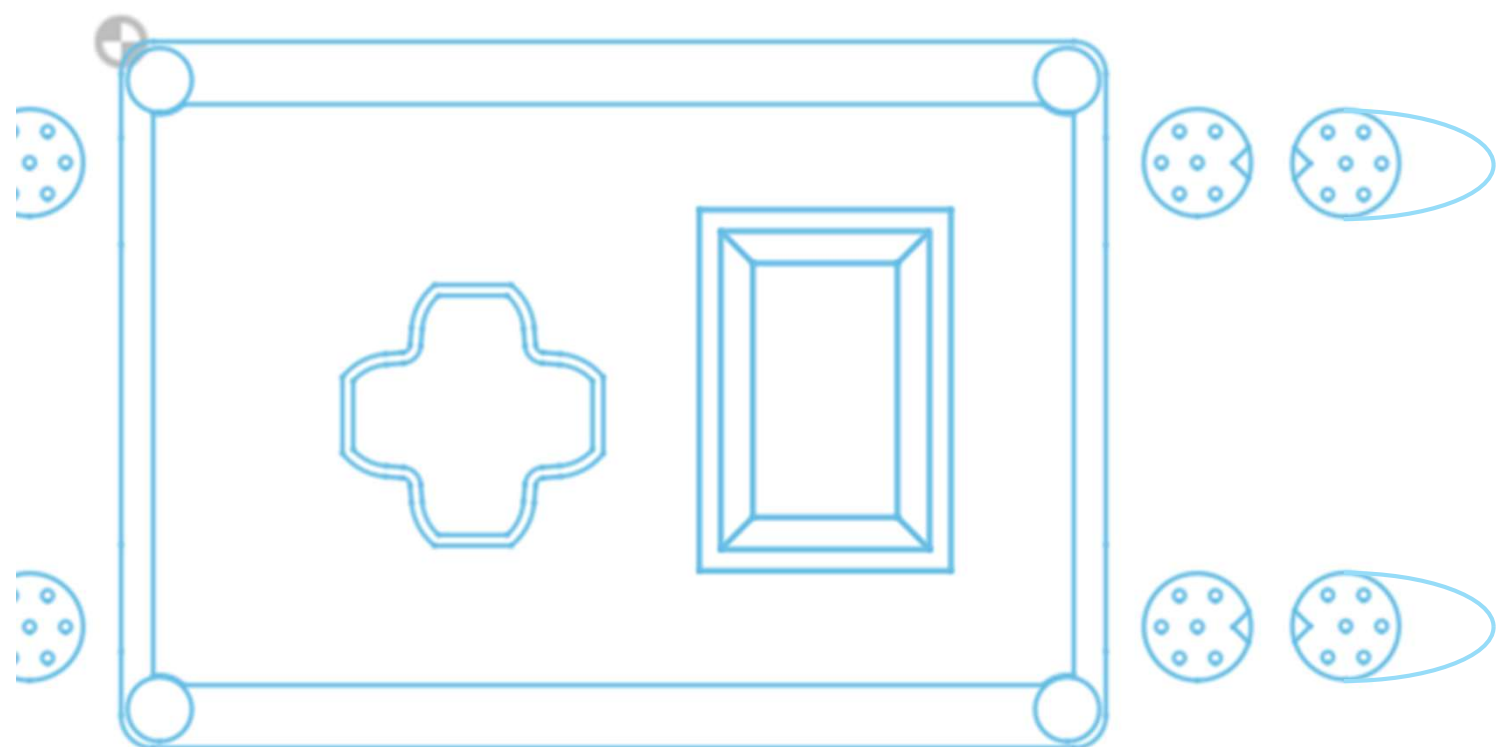


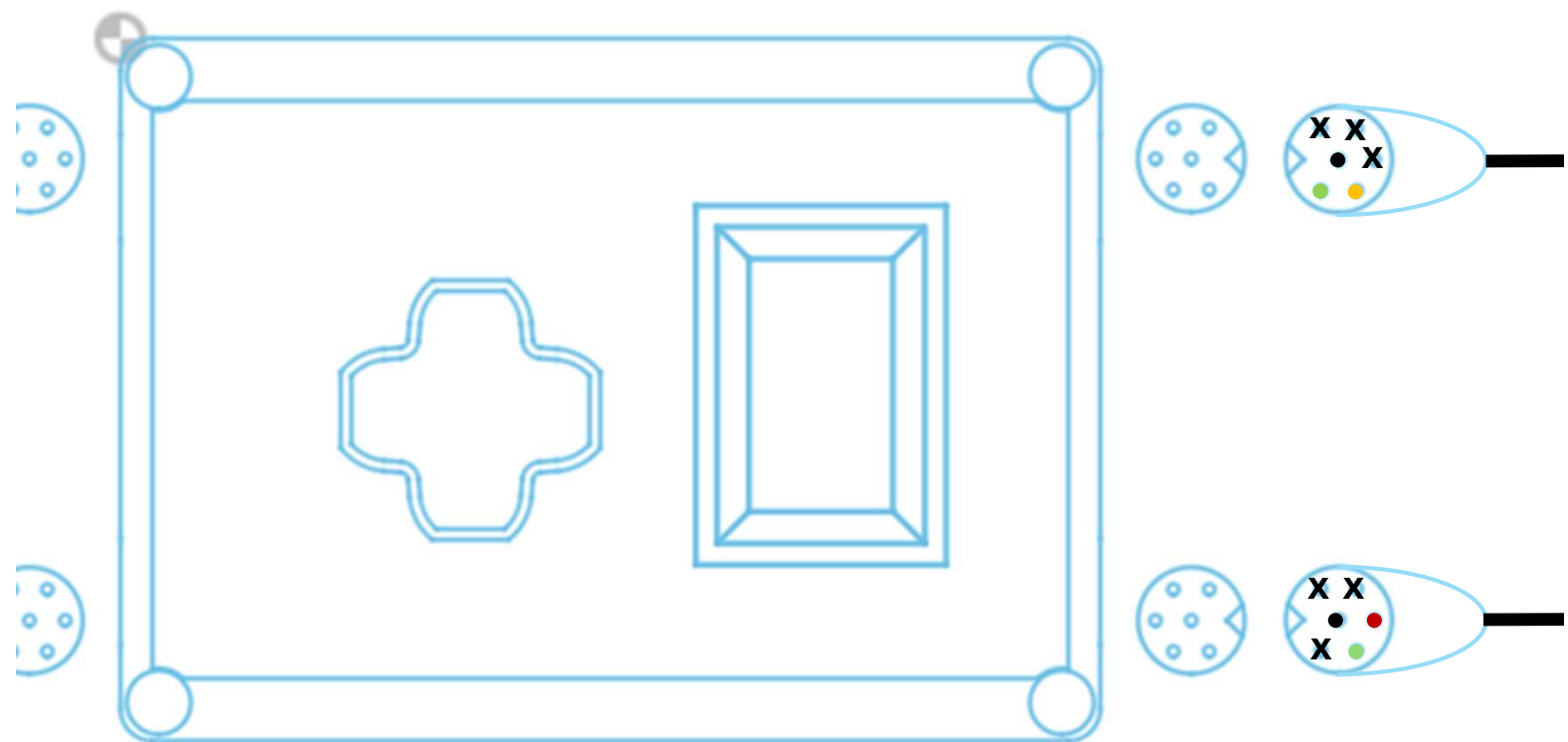
Logger 3V3

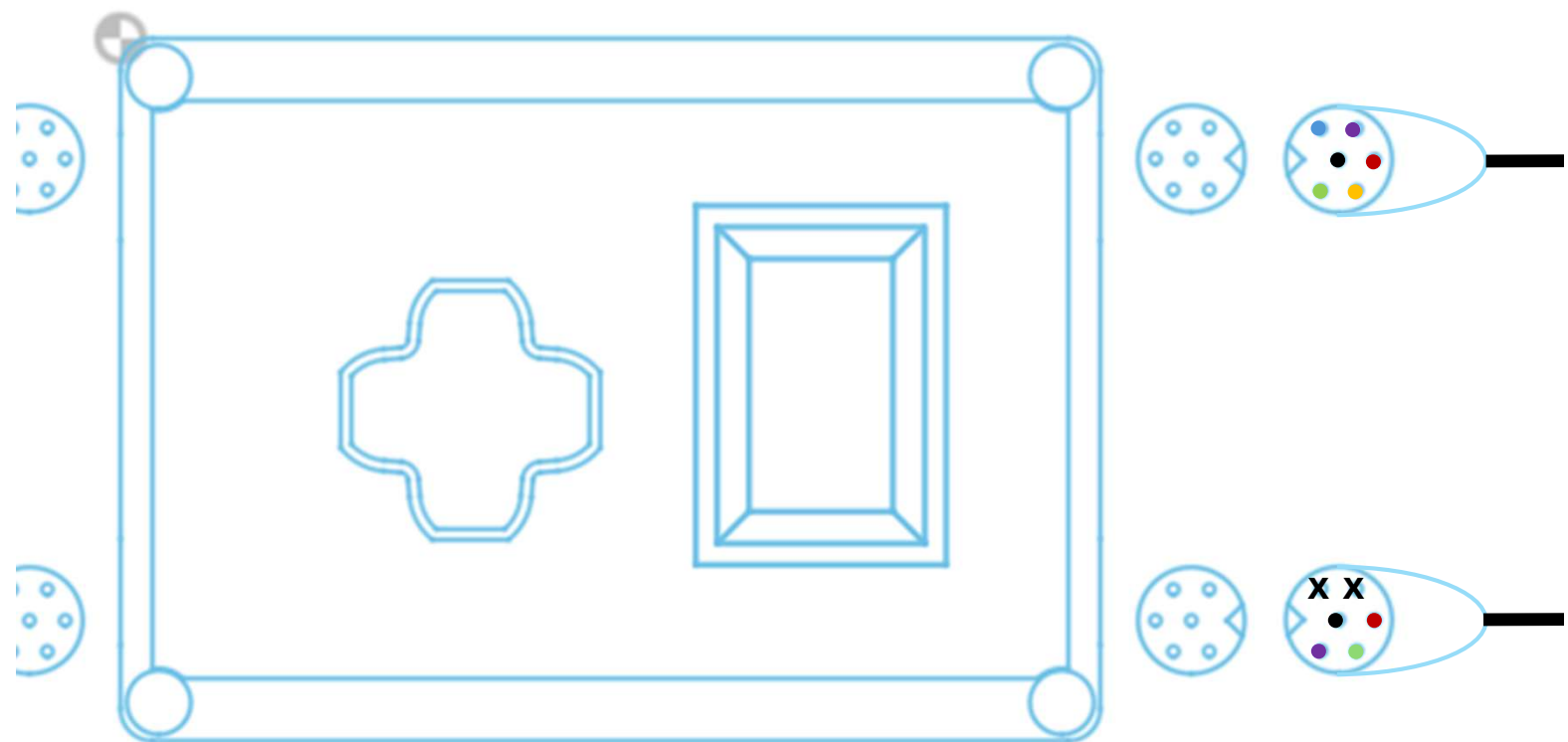
Logger SDA

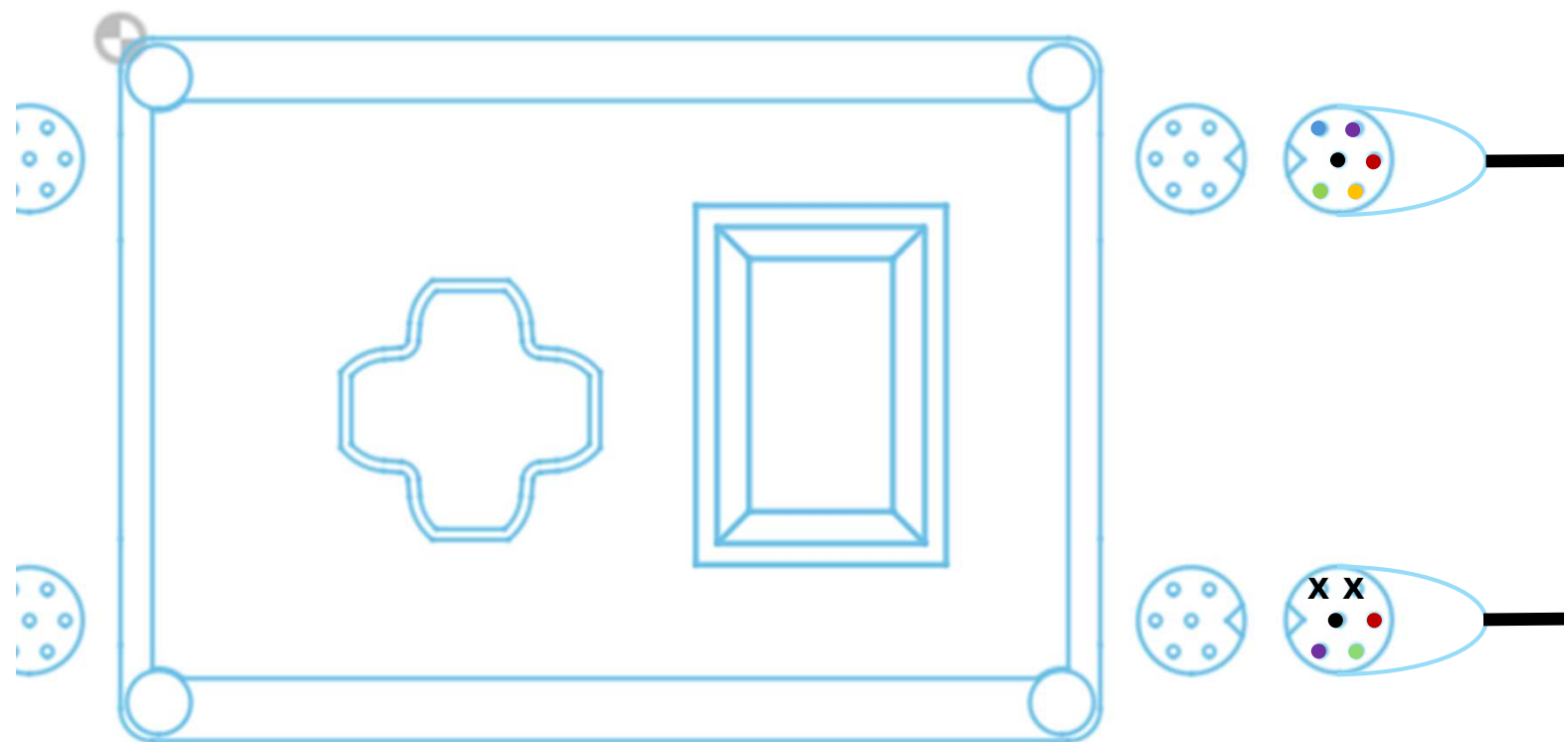
Logger SCL

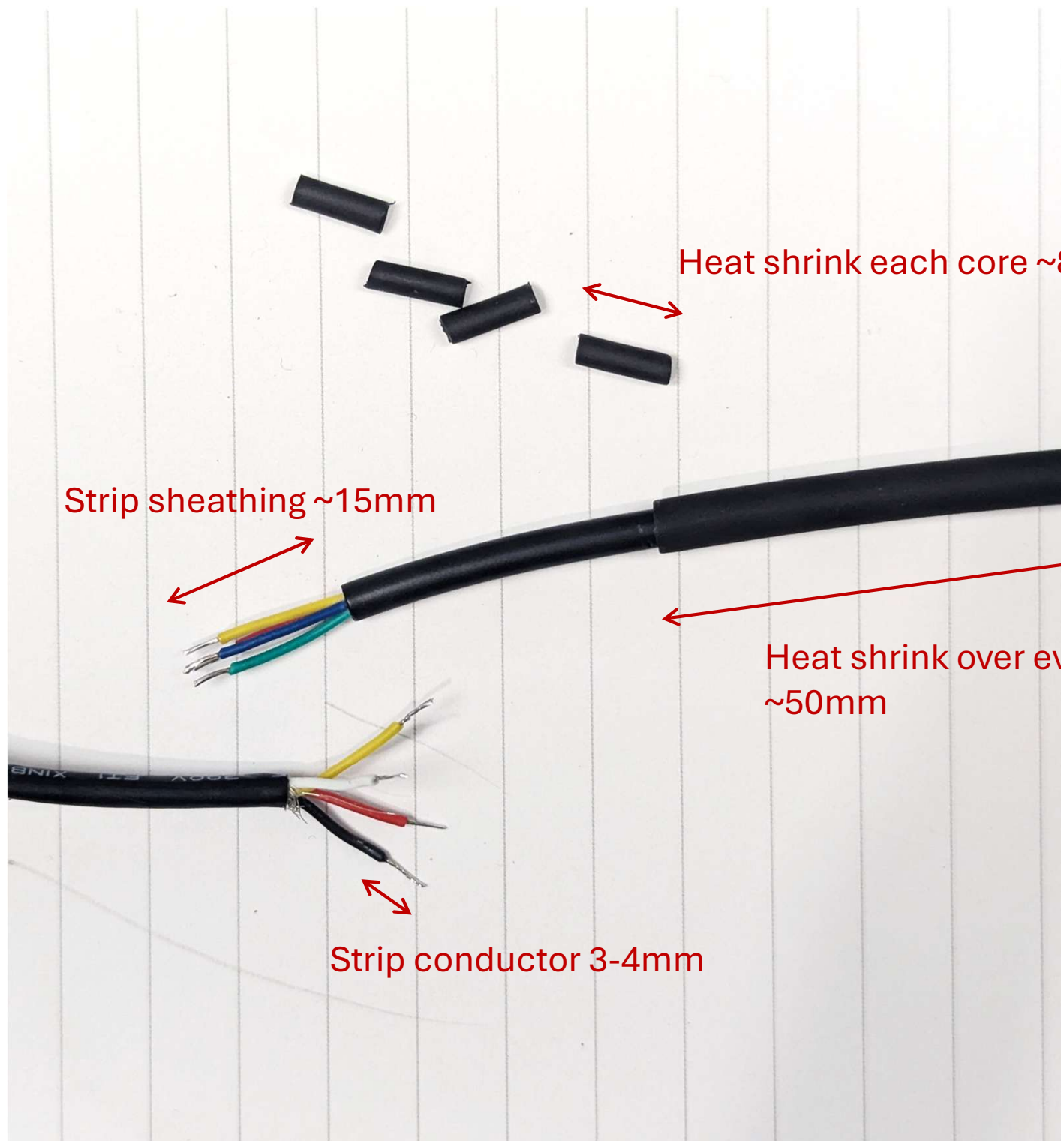
Logger GND





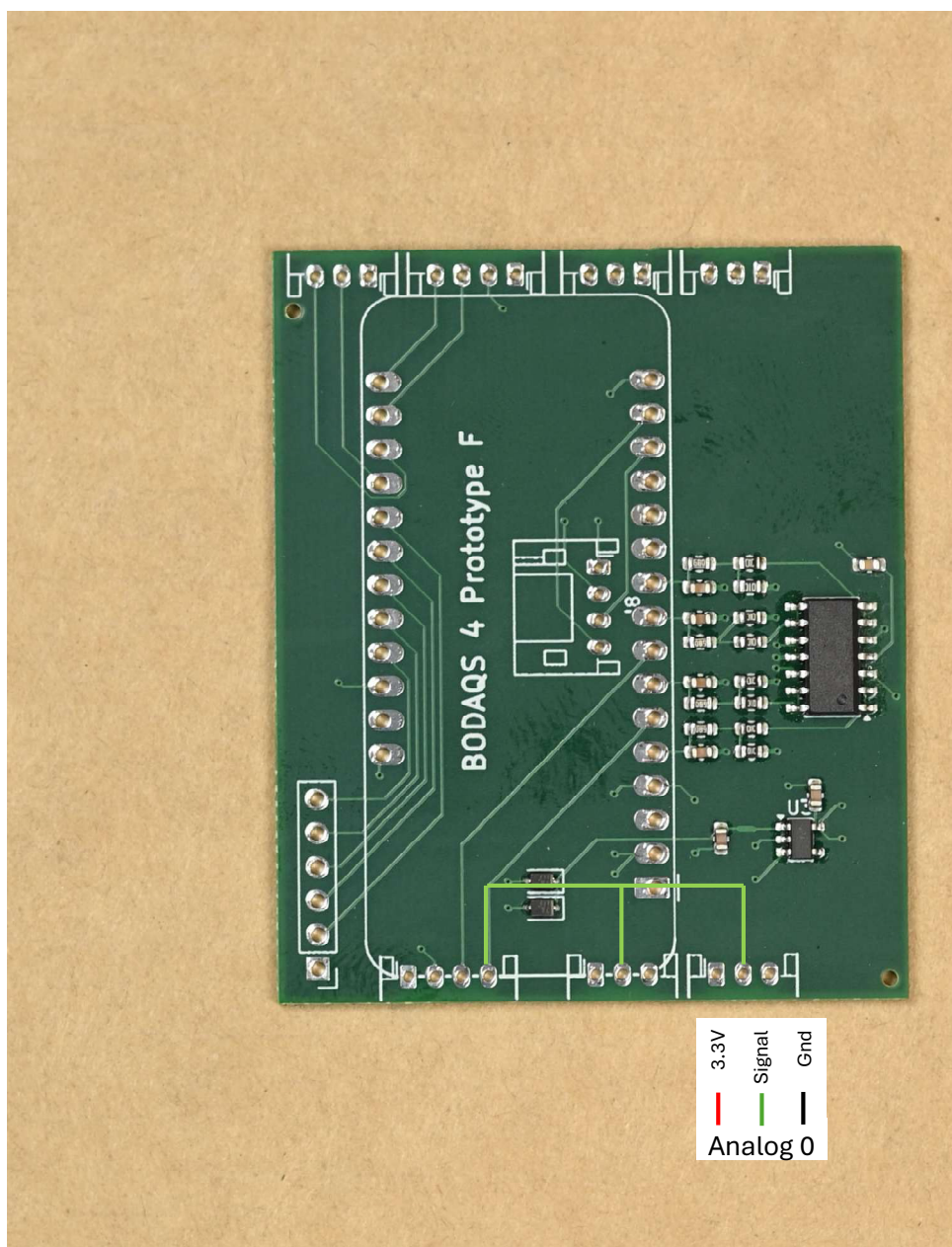




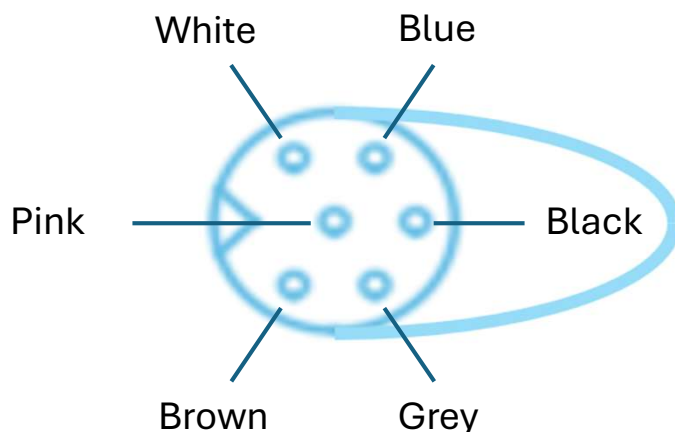


Analog 2

— 3.3V
— Signal
— Gnd
— SDA
— SCL



Wire color guide for Hysik Electronics 6-core signal cables, with A8 logger applications



Bar switch / UART

Pink	GND
White	TX
Blue	RX
Black	3V3
Grey	LED
Brown	Mark

ADC 0

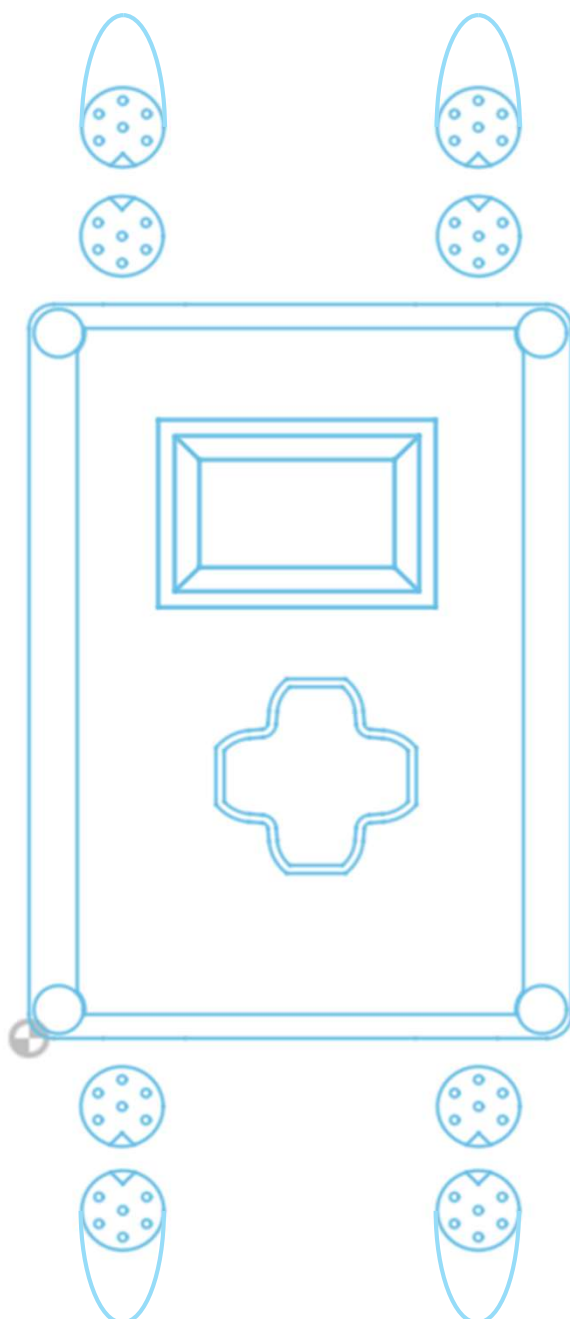
Pink	GND
White	A3
Blue	A2
Black	3V3
Grey	A1
Brown	A0

I2C

Pink	GND
White	SCL1
Blue	SDA0
Black	SCL0
Grey	3V3
Brown	SDA1

ADC 1

Pink	GND
White	A7
Blue	A6
Black	3V3
Grey	A5
Brown	A4





A8 Mountain Bike Data Logger

Handlebar
switch and GPS

ADC0
(4 analog inputs)



OLED display

Power switch

Navigation pad

```
Menu
> WiFi: OFF
  Upload: OFF
  Mute sensors
  Sample rate: 500 Hz
  Calibration
```

```
Menu
Sample rate: 500 Hz ^
Calibration
Sag helper
Sleep
> Settings
```

```
Settings
> WiFi mode: STA
  Log format
  Reset time
  Health
  Restart
```

I2C (2 digital
data buses)

ADC1
(4 analog inputs)

USB
charging

Key specifications

Inputs	8 analog; 2 I2C buses, 1 UART (typically for GPS), 1 handlebar switch
Sample rate	Up to 1000 Hz depending on sensor configuration
Communication	WiFi (2.4GHz), USB
Power	3.7V Li-Ion battery, USB-C charging
Output format	CSV, BODAQS Compact Binary
Configuration	Via web browser / on-device menu
Compatible software	BODAQS Desktop